Toward an optimal data-generating process for simulating predictive-biomarker interactions in aggregated N-of-1 trials

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Abstract

Background. A predictive biomarker is validated by an interaction test. In an aggregated N-of-1 design every participant is observed on and off drug, and the power of that test is estimated by simulation, which can only show what its data-generating process (DGP) allows. We set out three requirements for such a DGP: a valid (positive-definite) joint distribution, an interaction that survives carryover, and flexibility to impose stress conditions. We compare candidate choices for each of its elements against them. The DGP is a multivariate normal model in which the predictive biomarker is a correlation with the drug response (Hendrickson et al. 2020).

Methods. We derive the closed-form power of a paired-difference interaction statistic and a Schur-complement bound on the largest feasible interaction. We check both by Monte Carlo simulation.

We assume a biomarker that is predictive only: it is correlated with the drug response and not with baseline severity, the placebo response or the natural history. The Hybrid design is the primary case.

Results. The first element is the biomarker-response correlation at off-drug visits. If it stays constant after first exposure, power falls nearly to the test size under any carryover. If it decays at the carryover half-life, power returns to its no-carryover value at half-lives of 0.1 to 0.2 week and declines at longer ones. Second, the serial correlation structure bounds the largest interaction: 0.256 under compound symmetry and 0.474 under separable AR(1) at a serial correlation of 0.7. A PTSD symptom series suggests 0.9, where the bound falls to 0.34 and the Hybrid design needs about half the sample it needs at 0.7.

Conclusions. We recommend a DGP with a biomarker-response correlation that decays after discontinuation, a separable AR(1) structure, and a check that refuses infeasible effect sizes. We give closed-form results, code and planning sample sizes. Its parameter values are not yet calibrated, so it is a step toward an optimal DGP.

Trial registration. Not applicable.

Keywords: N-of-1 trials; predictive biomarker; simulation; carryover; data-generating process; power; prazosin.

Background

Validating a predictive biomarker requires a biomarker-by-treatment interaction test (1). In an aggregated N-of-1 design every participant is observed both on and off drug, so the test can be made within participants. Its power must be estimated by simulation.

A simulation can show only what its data-generating process (DGP) allows. The DGP decides how the biomarker acts, how the drug effect builds and wanes, and how one period carries into the next. Yet there is little guidance on what a DGP for these trials should be. In this paper we explore that question. We set out three requirements a DGP should meet. We then take its elements one at a time. For each element we compare candidate choices against the requirements and keep the better one. The result is a recommended DGP. We do not claim it is optimal. We say which of its parts remain uncalibrated.

The DGP builds on the model of Hendrickson et al. (2), the only DGP we found that encodes a predictive biomarker for these designs. It draws the baseline biomarker and three latent response components at every visit from one multivariate normal distribution. The trial design assembles the correlation matrix from each participant's exposure state. The biomarker enters as a correlation with the drug response. The primary case is the Hybrid design. It combines open-label titration, blinded discontinuation and a brief crossover, with carryover between phases.

Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times. The individual series are analyzed together (3–5). Every participant contributes both an on-drug and an off-drug response. The design can therefore estimate how the drug effect varies between people. It can then test whether a baseline measurement predicts who benefits (6,7). The motivating example is prazosin for post-traumatic stress disorder (PTSD). Prazosin has been tested in several randomized trials (8–10). The largest, the VA Cooperative Studies Program trial CSP 563 (304 veterans), found no benefit on its primary outcomes (10). In a secondary analysis of an earlier trial in 67 soldiers, higher pretreatment standing systolic blood pressure was associated with greater symptom reduction on prazosin but not on placebo (11). The difference was about 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg. Blood pressure was therefore proposed as a candidate predictive biomarker. That association did not replicate in our reanalysis of CSP 563 (Additional file 1; participant-level data not public). In 279 participants the arm-by-blood-pressure interaction was +0.25 points per 10 mm Hg (95% CI −2.72 to 3.21), which excludes the published value. Prazosin and blood pressure therefore serve here as a worked example for the units of the planning section, not as an established predictive biomarker.

Recent reviews cover the design and analysis of N-of-1 trials, including Bayesian models for series of trials (12,13). Closed-form power calculations exist only for simplified versions of these designs (14,15). Power is therefore estimated by simulating trials from a data-generating process (DGP) and analyzing each simulated trial (16). Several simulators of N-of-1 and aggregated N-of-1 data exist. Percha et al. (17) simulate single series with exponential wash-in and wash-out to choose block lengths and sampling rates. Chen and Chen (18) generate cycles under compound symmetry or AR(1) with a carryover effect to compare analyses. Blackston et al. (19) and Gärtner et al. (20) simulate aggregated series with carryover to compare designs and estimators. None of these

four represents a baseline biomarker that modifies the drug response. Tueller et al. (21) estimate power for idiographic within-subject trials by simulation. Grenet et al. (22) simulate crossover and parallel-group trials with a binary predictive biomarker. They encode it as a mean-term interaction. In single-case research, multilevel models with moderators of the intervention effect are established (23,24), and Moeyaert et al. (25) estimate by simulation the power to detect such a moderator. There, too, the moderator enters the mean model.

The correlation encoding

The DGP has three parts. The first is a component decomposition of the symptom score. The score is the baseline level minus three responses. Each response follows a modified Gompertz curve. The drug-response component (BR) is driven by time since the current on-drug period began. The placebo-response component (PB) is driven by time in the trial and scaled by the participant's expectancy, the probability of receiving drug given the design. The natural-history component (TV) is driven by time in the trial. The second part is the design specification for open-label, crossover, blinded-discontinuation and Hybrid designs. It tracks time on drug, time since discontinuation and expectancy at each visit. The third part is open software (the `pmsimstats` R package).

The DGP also encodes the predictive biomarker in a distinctive way. Statistical models and sample-size methods for N-of-1 and crossover designs usually represent treatment-effect heterogeneity as random participant-by-treatment variation (6,26–30). The DGP instead places the biomarker in the same multivariate normal distribution as the response components. The biomarker has a nonzero correlation c_{bm} with the drug response at on-drug visits. Without carryover it has zero correlation elsewhere. We call c_{bm} the biomarker-response correlation. A predictive biomarker, a moderator of the treatment effect (31), is then one whose association with the outcome differs between on-drug and off-drug visits. The parameterization keeps the marginal distribution of the drug response fixed. It expresses the strength of the interaction as a correlation. We call this the correlation encoding of the interaction.

The statement that the encoding has no precedent rests on three database searches. On 2026-10-27 we searched PubMed for (N-of-1 OR single patient OR aggregated N-of-1) AND simulation AND (biomarker OR effect modifier OR treatment-by-covariate) AND (power OR sample size). It returned one record, the study that proposed the encoding (2). On 2026-11-03 we searched PubMed titles and abstracts for ("N-of-1" OR "single-patient" OR "single-case" OR "aggregated N-of-1" OR

crossover) AND (simulation OR simulated OR "Monte Carlo") AND (biomarker OR moderator OR "effect modifier" OR "treatment-by-covariate") AND (power OR "sample size"). We ran the same terms, without "Monte Carlo", against abstracts in Europe PMC. Both returned two records: that study and a simulation of moderator power in single-case multilevel models (25). Neither search retrieved Grenet et al. (22), which we identified otherwise, so the searches are not exhaustive. Web of Science and Scopus were not searched. The searches cover the N-of-1, crossover and single-case literature indexed in these two databases. They do not cover the wider literature on joint simulation of covariates and outcomes, such as copula methods. In that literature, generating variables jointly from a specified correlation matrix, and checking that the matrix is feasible, is standard (32,33). What we found no precedent for is a biomarker-by-treatment interaction encoded as a correlation whose value depends on the design and the exposure state. The claim is therefore limited to the N-of-1 literature we searched.

Why the joint distribution is built from the design

The simplest DGP that could serve would generate one symptom score per visit. Its mean would carry the design effects and an interaction term $\beta\, B_i D_{it}$. Its residual would be compound symmetric or AR(1). Most simulators of crossover and N-of-1 data use that form (14,18), and the analysis model assumes it. It is adequate when the question is the main effect. For the questions this series asks, the joint-distribution form differs from it in three ways. The simple form stays usable: a mean model with a random participant-by-treatment slope (6,27) is the conditional equivalent of the joint-distribution form (Discussion).

1. **How the moderation persists after discontinuation is stated separately from the mean carryover.** Carryover in crossover trials is usually modeled as a mean effect (34), a model whose usefulness has been questioned (35,36). In the simple form the analyst adds it as a term such as $\beta B_i D_{bc,it}$. In the joint-distribution form it is the correlation profile g_t. Either is a modeling choice. The joint-distribution form makes the choice visible in the same quantity that sets the effect size.
2. **The interaction is expressed as a share of between-person variation.** With the marginal variance of the drug response held fixed, c_{bm}^2 is the share of the on-drug drug-response variance that the biomarker explains linearly. Appendix A.6 relates it to a responder-class model. A mean model with a random participant-by-treatment slope implies the same kind

of quantity through the ratio of the fixed slope to the slope variance. Neither form makes the residual variance depend on the biomarker.

3. **Existence becomes a question with an answer.** In the simple form every parameter combination yields a valid distribution. In the joint-distribution form, with the marginal variances fixed, it does not. Whether the simulated effect sizes are jointly possible, and how much of the interaction the design can see, then have closed-form answers.

In the joint-distribution form, the biomarker and the three response components at every visit form one multivariate normal vector. The design assembles its correlation matrix. The biomarker is correlated with the drug response at on-drug visits, and at a value set by the exposure state at off-drug visits. The components are serially correlated in calendar weeks. The components are cross-correlated through a common block. Three results follow. The largest feasible interaction has a closed form. It depends on the design and on two modeling choices: the marginal drug-response variance is held fixed, and the drug-response component keeps its full variance and serial correlation at off-drug and pre-exposure visits. We call it the feasibility bound. The power of a paired-difference statistic has a closed form. The design enters it through the on/off correlation difference, the on-drug correlation minus the mean off-drug correlation. The residual structure the analysis model should assume can be derived from the DGP. Three requirements come with this form. The marginal covariance of the outcome is implied rather than chosen. The joint distribution must exist. Every entry must be calibrated or declared uncalibrated. The Discussion shows that, conditional on the biomarker, the correlation encoding reduces to the mean-term form with a smaller residual variance, so the standard form remains available within it. We do not claim the correlation encoding is preferable to a mean-term or random-coefficient form.

The DGP is not new in its parts. Simulating repeated measures from a multivariate normal with compound symmetry or AR(1) and a carryover term is routine (14,18). Decomposing a clinical trajectory into disease progression, placebo response and drug action is the pharmacometric convention (37). Separable (Kronecker) covariances for several response variables measured repeatedly are an established model (38,39). Hendrickson et al. (2) combined these parts. It placed a baseline variable in the joint distribution of the latent response components, with a correlation indexed by each visit's exposure state. Designs for heterogeneity analyses in cluster randomized trials also use correlations of an effect modifier (40). These are intraclass correlations of the modifier and of the outcome, and the modification itself is a term in the mean model. This paper refines that

combination. We found no other instance of it in the N-of-1 literature we searched (above).

Requirements for the data-generating process

We judge candidate choices against three requirements. Each stage of the Results takes one element of the DGP, compares the candidates, and ends with the choice and the requirement it rests on.

1. **A valid joint distribution.** The biomarker, baseline and the three response components at every visit are drawn from one multivariate normal distribution. Their correlation matrix must therefore be positive definite: every eigenvalue must be positive. We meet this by construction in two steps. The response block uses a separable AR(1) structure, the Kronecker product of a cross-component matrix and an AR(1) matrix in calendar weeks. It is positive definite for any visit schedule (Appendix A.5). The biomarker row is then constrained. By the Schur complement, the full matrix is positive definite if and only if c_{bm} lies below a bound set by the design and the correlation profile, the feasibility bound (Appendix A.4). A matrix outside the bound is refused, not repaired.
2. **An interaction that survives carryover.** The biomarker-by-treatment interaction is a contrast between on-drug and off-drug visits, and under carryover it must remain one. The biomarker-response correlation off drug must therefore fall as the drug effect fades. We measure this by the on/off correlation difference, which governs power in closed form (Methods). The carryover half-life should mean the same thing for the drug-response mean and for the correlation.
3. **Flexibility for stress conditions.** The DGP should be able to impose the conditions under which analyses are compared, such as dropout, measurement error and a different decay form. These are options, off by default, so that their effect on size and power can be measured.

Where the requirements leave a choice open, we use pharmacological plausibility for a short-half-life, titrated drug such as prazosin, and say so where it decides a stage. Every parameter is labeled as a default value, a choice made here, or uncalibrated (Table 8). Calibration is reported, not required: the values have not been re-estimated for this paper, and a reanalysis of CSP 563 (planning section) estimates some of them on the total symptom score and shows that others cannot be identified from a parallel-group trial. A DGP meeting the three requirements is what we call the recommended DGP. Its label in the repository's code and data is `refined`.

The reference drug. Prazosin's plasma half-life is about 2.3 hours (41), against weekly visits. The plasma concentration therefore falls by orders of magnitude within a day of the last dose. We assume a symptom-level half-life of 0.1 to 0.2 week (17 to 34 hours), or 7 to 15 plasma half-lives, as the pharmacological range. No symptom-response half-life for prazosin in PTSD has been reported, so the range is an assumption. A drug's effect need not decline with its plasma concentration (42), so the plasma half-life is only a first guide. The Results therefore cover 0 to 2 weeks and label half-lives of 0.5 week and above as behavioral scenarios. The drug is re-titrated after an interruption, which bears on how the DGP treats re-exposure (Stage 3).

Aims

We have three aims. First, we judge the elements of a DGP for these trials against the three requirements, one element at a time. The elements are the handling of infeasible matrices, the biomarker-response correlation off drug, carryover on the mean, the serial correlation, the cross-component structure, and the optional features of measurement error, decay form and dropout. Second, we derive in closed form what each candidate does to feasibility and to power. Third, we collect the choices as a recommended DGP and give the sample sizes a trialist needs to plan under it. The Hybrid design is the primary case. The open-label, blinded-discontinuation and crossover designs are comparators.

The biomarker is assumed not to be prognostic. Throughout, the biomarker is predictive only. It is correlated with the drug response and with nothing else: its correlations with baseline severity, the placebo response and the natural history are zero. A biomarker that is also prognostic can bias the interaction test when the analysis model omits or misrepresents its prognostic effect (43,44). That case, the pathways through which it acts, their joint feasibility with the biomarker-response correlation, and the analyses that remain valid under it, are the subject of a companion paper (45). In the motivating trial the assumption is partly supported: in CSP 563 the biomarker is not associated with baseline severity (planning section). The data cannot show whether it is associated with the placebo response or the natural history.

Methods

The Methods describe the parts of the DGP that are fixed, the elements left open, and the tools used to compare candidate choices: the estimands, a statistic with a closed-form distribution, the feasibility bound and the simulation studies. Every choice about an open element is made in the Results.

The data-generating process

Participant i is observed at a baseline visit $t = 0$ and at post-baseline visits $t = 1, \dots, T$, at calendar weeks w_t (4, 8, 9, 10, 11, 12, 16 and 20 in the Hybrid design). The symptom score is

$$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}),$$

so a larger response lowers the score. Each component has a mean given by the modified Gompertz curve

$$G(x;\, m, d, r) = m\, \frac{\exp(-de^{-rx}) - e^{-d}}{1 - e^{-d}},$$

so that $G(0) = 0$ and $G(\infty) = m$. It is evaluated at a component-specific clock:

- The drug-response component (BR) at time on drug t_{od}, the time since the current on-drug period began, with $m = 10.99$, $d = 5$, $r = 0.42$ and SD 8. The curve reaches its inflection at 3.8 weeks, half its maximum at 4.7 weeks and 90% at 9.2 weeks. When the drug is resumed, time on drug restarts at zero.
- The placebo-response component (PB) at time in the trial, scaled by the expectancy η_t (1 in the open-label phase, 0.5 when blinded), with $m = 6.51$, $d = 5$, $r = 0.35$ and SD $10\eta_t$.
- The natural-history component (TV) at time in the trial, with the same curve as PB and SD 10.

The biomarker has mean 124.3 and SD 15.4, and baseline severity has mean 83.1 and SD 18.5. We call these the default values. They are the values of Hendrickson et al. (2) (their code at commit 58b32a9, stored in the package as `extracted_rp` and `extracted_bp`). Every simulation

and closed-form calculation in this paper uses them, including the simulations of the analysis G10 in the planning section and the paired rerun of Appendix B.4. In those two, the biomarker enters the analysis standardized to mean 0 and SD 1, so their interaction coefficients are per SD of biomarker. The standardization is linear and does not change the test.

The vector (biomarker, baseline, $TV_{i\cdot}$, $PB_{i\cdot}$, $BR_{i\cdot}$) is multivariate normal. The biomarker's correlation with BR_{it} is $c_{bm}\, g_t$, where g_t is the correlation profile, the fraction of c_{bm} present at visit t. Its correlations with baseline, TV and PB are zero (Aims). Baseline is uncorrelated with the response components. Participants are assigned to the design's paths in fixed numbers (18, 18, 17 and 17 in the Hybrid design at $N = 70$).

The open elements. Five elements are decided in the Results, each by comparing candidates:

1. *Infeasible matrices* (Stage 1): when the correlation matrix is not positive definite, replace it with a nearby valid matrix, or refuse it.
2. *The correlation profile off drug* (Stage 2): the constant rule sets $g_t = 1$ wherever the drug-response mean is nonzero, so the correlation is constant after first exposure; the decaying rule sets $g_t = 2^{-t_{sd}/t_{1/2}}$ off drug after exposure, with t_{sd} the time since discontinuation and $t_{1/2}$ the carryover half-life.
3. *The drug-response mean off drug* (Stage 3): the compounding rule decays each off-drug mean from the previous visit's, $\mu_{BR,t} = \mu_{BR,t-1}\, 2^{-t_{sd,t}/t_{1/2}}$; the anchored rule decays it from the last on-drug mean.
4. *The serial correlation within a component* (Stage 4): compound symmetry, the same correlation ρ_{cs} between any two visits, or AR(1) in calendar weeks, $\rho^{|w_t - w_s|}$. We write ρ_{cs} to keep ρ for the AR(1) correlation per week.
5. *The correlation between components* (Stage 5): c_1 at the same visit and a constant $c_\times$ otherwise; c_1 at the same visit and $c_\times \rho^{|w_t - w_s|}$ otherwise; or the separable form $c_1 \rho^{|w_t - w_s|}$ at every lag.

The binary-indicator analysis is the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T w_t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it},$$

with a random intercept u_i, the binary drug indicator D_{it}, and baseline included as a row with $D_{i0} = 0$. The interaction $\beta_{bm:D}$ is tested with Satterthwaite degrees of freedom (46).

Two standard errors. Every mixed model in this paper assumes a covariance for each participant's outcomes, its working covariance. The *working-covariance standard error* is the model-based one, computed as if that covariance were the true covariance of the outcomes. It is correct when the working covariance is right and can be too small or too large when it is not. The *cluster-robust standard error* treats each participant as a cluster and estimates the variance of the fixed effects from the residuals (the sandwich estimator), so it remains valid when the working covariance is wrong. We use its bias-reduced form, CR2, with Satterthwaite degrees of freedom (47,48). The binary-indicator analysis uses the working-covariance standard error.

Estimands

The target is the on-drug biomarker moderation of the drug response. This is the change in the drug effect per unit of biomarker while the drug is fully present. Without carryover it equals $\beta_{bm:D} = -c_{bm}\sigma_{BR}/\sigma_{bm}$.

Analyses that contrast on-drug with off-drug visits through a binary drug indicator estimate this quantity only when no moderation remains off drug. Under carryover they estimate an attenuated on-minus-off contrast. Analyses with an exposure-decayed indicator $D_{bc,it}$ at the true half-life recover the on-drug moderation, because the decaying rule and $D_{bc,it}$ share one exponential decay. Under the Weibull decay option, or with a correlation half-life that differs from the carryover half-life, they do not. At another analysis half-life, fixed or chosen from the data, they recover it only approximately. Table 1 states which analysis in this paper targets which quantity. The three sets of results are compared only where their targets coincide (no carryover) or where the attenuation is the object of study (Stage 2).

Table 1: The analyses used in this paper and the quantity each estimates.

Analysis	Drug indicator	Estimates	Used in
Paired-difference statistic E9 (closed form)	binary	on-minus-off contrast, $\beta_{bm:D}$ times the on/off correlation difference	Stages 1, 2, 4, 5; planning

Analysis	Drug indicator	Estimates	Used in
The binary-indicator analysis (mixed model)	binary	the same contrast	Study 1; Differences from Hendrickson et al.
Study 2: the binary-indicator analysis	binary	the on-minus-off contrast	Stage 6
Study 2: fixed-half-life analyses	$D_{bc,it}$ at an analysis half-life of 0.5 week, the true half-life except in the cells that vary it	on-drug moderation $\beta_{bm:D}$ where the half-lives match	Stage 6
Study 2: analyses with the half-life chosen by AIC	$D_{bc,it}$ at the AIC-chosen half-life (0, 0.25, 0.5, 1 or 2 weeks)	on-drug moderation $\beta_{bm:D}$ where the chosen half-life is the true one	Stage 6
G10 (planning)	$D_{bc,it}$ at the half-life chosen by AIC without the interaction (0.25, 0.5, 1 or 2 weeks)	on-drug moderation $\beta_{bm:D}$ where the chosen half-life is the true one	Stage 2; planning

The paired-difference statistic

To isolate how carryover acts, we use a summary statistic whose distribution can be derived exactly (Appendix A). For each participant,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it},$$

the difference between the mean on-drug and mean off-drug scores. It is regressed on B_i by least squares. We call it E9 when it uses post-baseline visits only; E9 is the statistic used throughout. A variant, E9b, counts baseline as one more off-drug visit, as the binary-indicator analysis does. It is used only to explain the results of the binary-indicator analysis (section "Differences from the DGP of Hendrickson et al."). In the Hybrid design, Monte Carlo simulation of the binary-indicator mixed model shows that E9b reproduces its mean estimates to within 0.004. It reproduces its power closely (0.654 against 0.632 without carryover). In the crossover design the closed form understates the mixed model's power (0.645 against 0.708 without carryover, and 0.279 against 0.31 to 0.35 under the constant rule). The mixed model's time term uses the between-path contrast that the paired difference discards. The derivation gives three results.

- **Proposition 1.** The variance of Δ does not depend on the carryover half-life. The half-life enters only through the response means and the correlation profile. Neither appears in the variance.
- **Proposition 2.** The population slope is

$$\beta_{bm:D}^{\text{E9}} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\bar{g}_{\text{off}}\right), \qquad \beta_{bm:D}^{\text{E9b}} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\frac{n_{\text{off}}}{n_{\text{off}}+1}\,\bar{g}_{\text{off}}\right),$$

  where $\bar{g}_{\text{off}}$ is the mean of g_t over the n_{off} post-baseline off-drug visits. The slopes are specific to each path. In the Hybrid design n_{off} is 3 in two paths and 4 in the other two. The pooled slope is their allocation-weighted average.
- **Power** is a noncentral-t probability governed by the correlation $r_{\Delta B}$ between Δ and B. For E9 that correlation is proportional to the on/off correlation difference $\mathcal{G} = 1 - \bar{g}_{\text{off}}$.

So carryover can affect power only through the on/off correlation difference. The propositions reduce the question of why power falls to the question of what the correlation rule does to $\bar{g}_{\text{off}}$.

The feasibility bound

Write the response block of the correlation matrix as M and the biomarker's correlations with the response components as $c_{bm}\tilde{g}$, with $\tilde{g}$ the correlation profile padded with zeros at the TV and PB

entries. The full matrix is positive definite if and only if M is, and

$$c_{bm}^2 \, \tilde{g}^\top M^{-1} \tilde{g} < 1.$$

The left side is the squared multiple correlation of the biomarker on the whole response trajectory. The largest feasible c_{bm}, the feasibility bound, is therefore $c_{bm}^* = (\tilde{g}^\top M^{-1} \tilde{g})^{-1/2}$ (Appendix A.4). It depends on the correlation structure, the design and the correlation profile. It is a bound on the DGP, not on any analysis.

Simulation studies

We designed and report the simulations following the ADEMP structure of Morris et al. (16,49): aims, data-generating mechanisms, estimands, methods and performance measures. There are two studies (Table 2). The scripts are listed under Availability of data and materials. The closed-form calculations of Stages 4 and 5 and of the planning section use the same code as study 1 with ρ varied. A further run, described in the planning section, simulates the analysis G10 of the companion paper (50) under the recommended DGP with the default values. It supplies the G10 columns of the planning tables, the G10 sample sizes for 80 percent power and the G10 panel of Figure 4B (Figure 3). Its generator reproduces the closed-form power of E9 within two Monte Carlo SEs in the Hybrid, OL+BDC and crossover designs ($N = 70$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$ week, 1,000 replicates; `42-check-param-sets.R`).

Table 2: Design of the two simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for R replicates.

	1. Closed-form verification	2. Stress test of the optional features
Aim	Check the closed-form moments and power of Appendix A, and the loss of power under the constant rule	Find what the optional features of the recommended DGP do to size and power

	1. Closed-form verification	2. Stress test of the optional features
Data-generating mechanisms	The correlation encoding with the constant rule and with the decaying rule; compound symmetry and the separable AR(1) form; carryover on the drug-response mean decaying from the last on-drug value; Hybrid, crossover and OL+BDC designs; $N = 70$; grids of c_{bm} and $t_{1/2}$	The recommended DGP (Table 8), run three times: the reported run as defined, the first with the compounding carryover rule and the second with randomized allocation; base cell separable AR(1), $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week; 11 null cells and alternative cells: power curves under the separable form (c_{bm} from -0.30 to 0.45, 7 cells) and under compound symmetry ($\rho_{cs} = 0.8$, c_{bm} from -0.20 to 0.25, 6 cells), and cells at $c_{bm} = 0.30$ that each vary one feature (N, half-life, decay form, dropout or measurement error). The study also ran cells with a prognostic biomarker; they are reported in the companion paper
Estimand	$\beta_{bm:D}$, and the mean of the paired-difference statistic	$\beta_{bm:D}$
Methods	Paired-difference statistics E9 and E9b, closed form and simulated; the binary-indicator mixed model fitted with `lmer`	Fourteen analyses fitted to the same simulated trials, of which ten are reported; two-sided tests at $\alpha = 0.05$
Performance measures	Mean, SD and power of the statistic; agreement as (simulated − closed form)/MCSE	Null rejection rate; power; failed fits
Replicates	5,000 per cell (MCSE of power at most 0.007); 500 to 1,000 for `lmer`	1,000 per null cell (MCSE 0.0069 at a rate of 0.05), 500 per alternative cell (at most 0.022)

Number of replicates. Replicate counts were set by the precision each comparison needs. In study 2, 1,000 null replicates give a two-MCSE band of 0.036 to 0.064 for a single cell. Pooling the 11 null cells gives the band of 0.046 to 0.054 used in Stage 6. In study 1, a cell is flagged when the simulated and closed-form values differ by more than 2 MCSE. Chance alone produces this in about 4.6% of cells, or about 26 of the 566 cell-variants (three designs, three correlation structures, two statistics); 16 were flagged. The cell-variants are not independent, since each uses its own draws but the same designs.

Random numbers. Every cell, or chunk of a cell, sets its own seed from a fixed rule. In study 1 the seed is a function of the cell's parameters. Study 2 uses $20261007 + 1000 \times \text{cell} + \text{chunk}$. Results therefore do not depend on the number of cores or the order in which chunks run. All analyses in study 2 are fitted to the same trials. The comparisons are therefore paired. R's default Mersenne-Twister generator was used throughout.

Failed fits. In study 2, a fit that returned an error was recorded as missing. In the reported run, failures numbered 0 to 10 per analysis over 28,000 trials, and at most 0.3% in any cell. In the first and second runs they numbered 0 to 12 and 0 to 7, and at most 0.6% and 0.4% in any cell. Study 1 does not fit models for the paired-difference statistic, so it has no failures.

Which form of the DGP each study used. Study 1 and the closed-form calculations build the cross-component correlation in one of two ways. Where the separable structure is named, it is $c_1 \rho^{|w_t - w_s|}$. Where compound symmetry is named, it is $\rho_{cs} = 0.8$ within components, with $c_1 = 0.2$ at the same visit and $c_\times = 0.1$ otherwise between components. Study 2 was run three times, with the separable form in every cell except the compound-symmetry curve and an analysis half-life of 0.5 week. The first run used fixed path allocation (18, 18, 17, 17) and the compounding carryover rule on the mean. The second run used the anchored carryover rule, the rule of the recommended DGP, and also assigned each participant to a path at random in every trial. Randomized allocation is a driver option, not part of the recommended DGP as defined (Table 8). The third run is the recommended DGP as defined: the anchored carryover rule with fixed allocation. It differs from the first run only in the carryover rule and from the second only in the allocation, so the two effects can be separated. The tables in Stage 6 are from the third run.

Software. R 4.6.1, with lme4 2.0.1 and lmerTest 3.2.1 for the binary-indicator mixed model of study 1, nlme 3.1-169, mmrm 0.3.17, clubSandwich 0.7.0 and MASS 7.3-65. In study 2 the binary-indicator analysis is fitted with `mmrm` and a compound-symmetry covariance with Satterthwaite

inference, not with `lmer`. The two coincide when the within-person correlation is non-negative.

Exploratory status. The stress test is exploratory. Its cells and analyses were written down as a working protocol (Additional file 2). No analysis plan was fixed in advance, and the protocol was revised as results came in. Cells and three analyses were added after the first results. Failed fits were first counted as non-rejections and are now excluded from the denominator, a change of at most 0.0025 in any rate. The study also compared the analyses' power and selected one by a decision rule. That selection was revised after results, is post hoc, and is reported in Additional file 3, not here. Such post hoc changes can favor one method in comparative simulation studies (51), which is a further reason not to rank the analyses here. This paper uses the stress test only for what it shows about the DGP: what the optional features do to size and power.

Results

The Results are organized as stages. Each stage takes one element of the DGP. It states the candidate choices and the evidence. It ends with the choice made and the requirement it rests on. Table 8, at the end, collects the choices.

Stage 1. Feasibility of the joint distribution

The first question is whether the joint distribution exists for the effect sizes a study simulates. Under compound symmetry the correlation matrix is positive definite only for c_{bm} below a bound. Without carryover the bound is 0.256 in the Hybrid design. It is 0.256 in the crossover and 0.262 in the open-label-plus-blinded-discontinuation design. These values follow from the feasibility bound with M the compound-symmetry response block ($\rho_{cs} = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$).

But why is the bound so low? It is the condition that the biomarker's squared multiple correlation on the whole response trajectory stays below one. Under compound symmetry every visit's drug response shares the same person-level factor. The biomarker therefore cannot be correlated with the on-drug visits alone. Through that factor it is correlated with the off-drug visits as well. The bound is low for two reasons. The correlation profile switches between visits that the structure ties together. The drug-response component also keeps its full variance at off-drug visits. Stages 4 and 5 show how far a serial correlation that decays with lag raises the bound. Even then, an effect size of 0.6 is infeasible in 75% of the grid of Table 14 and in every Hybrid cell.

Table 3: Closed-form power of the paired-difference statistic E9 by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry. Constant: the constant rule (constant correlation after first exposure). Decaying: the decaying rule (correlation decaying at the carryover half-life). Against Monte Carlo estimates (5,000 replicates per cell), 0 of the 12 cells in this table differ by more than 2 Monte Carlo SE, and 16 of all 566 cell-variants of study 1 (about 26 expected by chance).

Correlation rule	0	0.1	0.2	0.5	1	2
Constant	0.721	0.050	0.050	0.050	0.050	0.050
Decaying	0.721	0.721	0.712	0.631	0.473	0.278

When a matrix is infeasible, a DGP can either repair it or refuse it. A repair replaces it with a nearby valid matrix, for example Higham's nearest positive semidefinite matrix (52) (Appendix B.2). A repaired matrix simulates a different DGP from the one stated, and it does not report which correlation changed. The recommended DGP therefore stops and reports the cell as infeasible. The package function `buildSigma()` does so by default (Appendix B.5). The choice rests on requirement 1, a valid joint distribution. What the bound means in outcome units, and what to do above it, are taken up in the planning section and the Discussion.

Stage 2. The decaying biomarker-response correlation

This stage takes the biomarker-response correlation at off-drug visits. There are two candidates: the constant rule and the decaying rule (Methods).

The constant rule. Under the constant rule every off-drug visit in the Hybrid design follows an exposure. With any positive half-life its drug-response mean is positive, so $g_t = 1$. Hence $\bar{g}_{\text{off}} = 1$ and, by Proposition 2, $\beta^{\text{E9}}_{bm:D} = 0$ for every $t_{1/2} > 0$.

Power is a step function of the half-life (Table 3, Figure 1). It takes its no-carryover value at $t_{1/2} = 0$. It equals the test size, 0.050, at every positive half-life of practical interest, whether 0.1 week or 10. At 0.1 week only 0.1% of the drug effect remains a week after stopping, yet the off-drug correlation is the full on-drug value. (Below about 0.007 week the computed drug-response mean underflows to zero in double-precision arithmetic, and the rule then behaves as if there were no carryover. This is an artifact of the computation, not a property of the DGP.) With a constant correlation, the biomarker moderates the on-drug and post-baseline off-drug responses equally. No interaction remains among the post-baseline visits, and E9 power is the test size. The loss of power under small carryover is therefore a property of the correlation rule, not of carryover or of the designs.

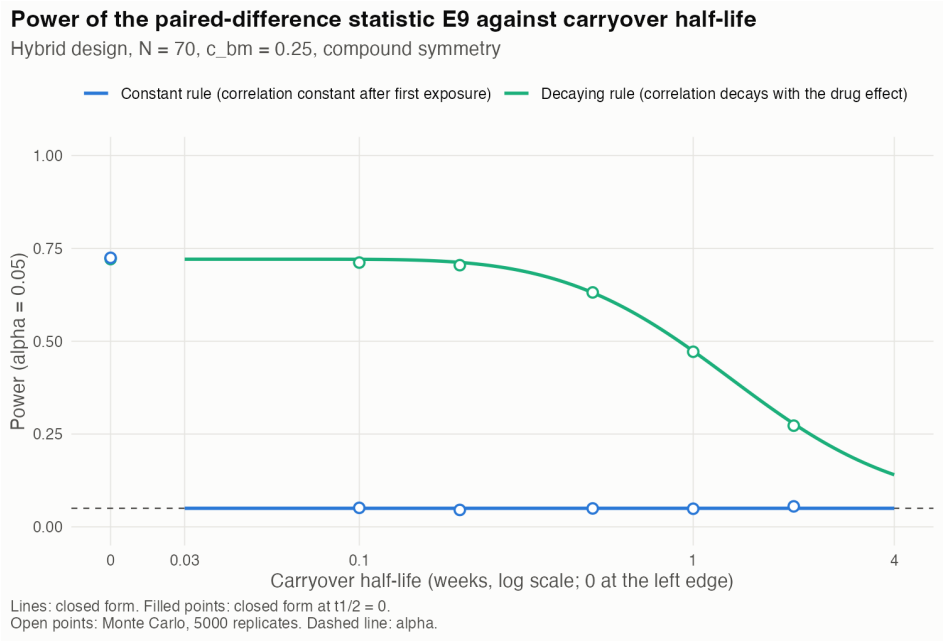


Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points; 5,000 replicates per cell), under the constant rule (constant correlation after first exposure) and the decaying rule (correlation decaying at the carryover half-life). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry; statistic E9.

The decaying rule. The alternative is a correlation that decays after discontinuation at the carryover half-life: $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}}$ off drug after exposure, and 0 before any exposure. This is the pharmacological reading of a predictive biomarker. The biomarker predicts the size of the drug effect, so it cannot predict a drug effect that is no longer present. Under the decaying rule the mean of g_t over the off-drug visits is

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t \,\text{off}} 2^{-t_{sd,t}/t_{1/2}},$$

which is strictly increasing in $t_{1/2}$, because every off-drug visit has $t_{sd} \geq 1$ week. Two consequences follow.

- **Power is strictly decreasing in the half-life** for the path-stratified paired-difference statistic. This holds on every path with at least one off-drug visit after an exposure (Proposition 3, Appendix A.3). The placebo-first crossover path, whose off-drug visits all precede exposure, has constant power. For the pooled statistic power is non-increasing at every step of a 120-point grid from 0.03 to 4 weeks.

- **The loss vanishes faster than any power of the half-life as $t_{1/2} \to 0$.** At 0.1 week, $\bar{g}_{\text{off}} \leq 2^{-10}$ and power is unchanged to within 0.001.

The constant rule is the limit of the decaying rule as $t_{1/2} \to \infty$, applied at every positive half-life.

Power depends on the correlation profile only through the on/off correlation difference $\mathcal{G} = 1 - \bar{g}_{\text{off}}$ (Figure 2). The difference is 1 without carryover. Under the decaying rule in the Hybrid design it is 0.9997 at 0.1 week, 0.991 at 0.2, 0.906 at 0.5, 0.754 at 1 and 0.551 at 2. Under the constant rule it is 0 at every positive half-life. The noncentrality is proportional to $r_{\Delta B}/\sqrt{1 - r_{\Delta B}^2}$, and $r_{\Delta B}$ is proportional to the difference. For small $r_{\Delta B}$ the sample size needed for a given power therefore scales approximately as $1/\mathcal{G}^2$. At 1 week this is about 1.8 times the no-carryover sample. At 2 weeks it is about 3.3 times.

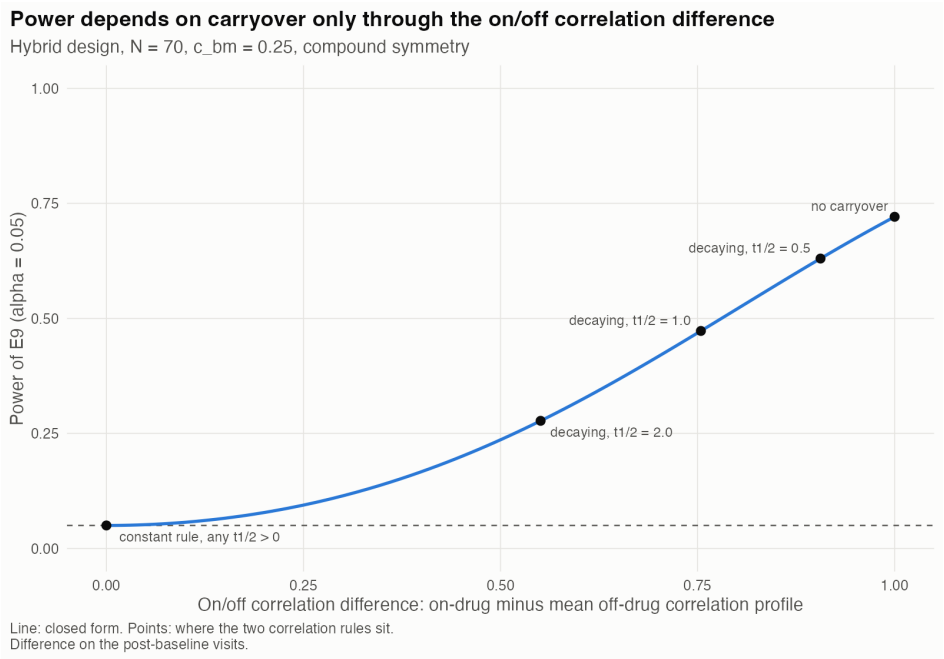


Figure 2: Closed-form power against the on/off correlation difference, with the positions of the two correlation rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$, E9.

Figure 3 shows the power of a mixed-model analysis under the recommended DGP of Stage 7, across four designs, two sample sizes, five censoring patterns and three half-lives. The analysis is G10, which the companion paper recommends (50). G10 is a random-intercept model with CAR(1) residual correlation and CR2 cluster-robust standard errors. Its exposure-decayed drug indicator has a half-life chosen by AIC on the model without the interaction (planning section). Each cell has 250 replicates and a Monte Carlo SE of at most 0.031. The open-label design is not estimable under

G10: every visit is on drug, so the exposure-decayed indicator is constant at 1 and the interaction term is collinear with the biomarker. The binary-indicator analysis replaces the interaction with a biomarker-by-time term in that case. G10 has no such substitute, and the cells are marked as not estimable rather than simulated. In the censoring patterns, dropout rises with worsening (Appendix B.5). Results at $N = 70$ without censoring:

- **Hybrid design.** Power is 0.34, 0.38 and 0.34 at half-lives of 0, 0.1 and 0.2 week.
- **Open label plus blinded discontinuation.** Power is 0.34, 0.28 and 0.24.
- **Crossover.** Power is 0.20, 0.19 and 0.27.

We compare G10 power at the pharmacological half-lives with its no-carryover value. Two rates differ when they are more than two combined Monte Carlo SEs apart. In the Hybrid design, power is within Monte Carlo error of its no-carryover value at both half-lives. In the OL+BDC design it is within Monte Carlo error of its no-carryover value at 0.1 week and lower (0.24) at 0.2 week. In the crossover design it is within Monte Carlo error of its no-carryover value at both half-lives. Each comparison rests on one cell of 250 replicates. In the planning run of the same DGP and analysis, with 1,000 replicates per cell (planning section), OL+BDC power is 0.295 without carryover and 0.275 at 0.2 week, within Monte Carlo error of each other. Of the censoring patterns, the 'high dropout' pattern gives the lowest power in the Hybrid design at 0.2 week, 0.14.

Figure 4 extends the comparison over interaction strengths and longer half-lives. It shows closed-form power in the Hybrid design for each correlation rule.

- **The constant rule.** Every column after $t_{1/2} = 0$ falls to the test size, at every strength. For example, at $c_{bm} = 0.2$ power falls from 0.52 to 0.05.
- **The decaying rule.** Power declines smoothly: 0.52, 0.52, 0.51, 0.44, 0.32 and 0.19 at the same strength over half-lives of 0 to 2 weeks.
- **Feasibility.** Gray cells mark effect sizes at or above the compound-symmetry feasibility bound, where no joint distribution exists. Under the decaying rule, $c_{bm} = 0.3$ becomes feasible only from $t_{1/2} = 1$ week. A correlation that decays instead of switching places a smaller demand on the structure (Appendix A.5).

The recommended DGP uses the decaying biomarker-response correlation. The choice rests on requirement 2, an interaction that survives carryover. Pharmacological plausibility supports it:

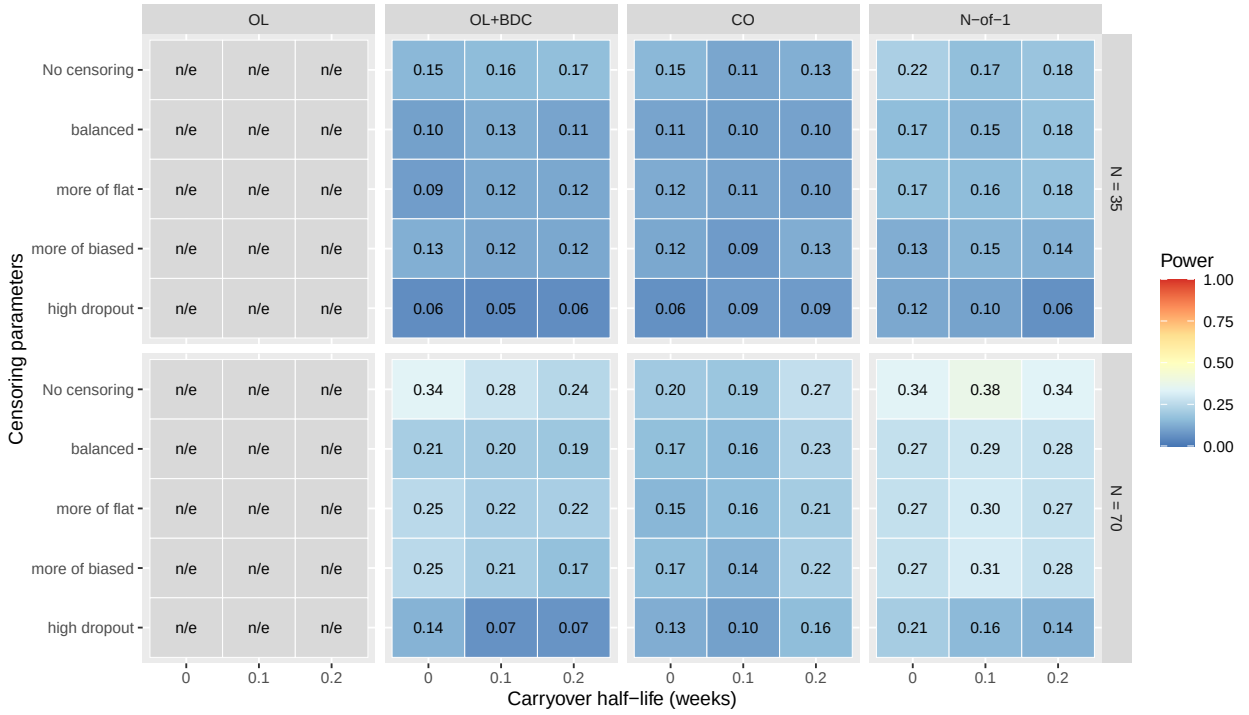


Figure 3: Power of the analysis G10 to detect the biomarker-by-treatment interaction under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying biomarker-response correlation, carryover decaying from the last on-drug mean, default values). Rows within panels: censoring patterns, under which dropout rises with worsening; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.031). Gray cells (n/e): the open-label design, in which every visit is on drug and the interaction is not estimable under G10.

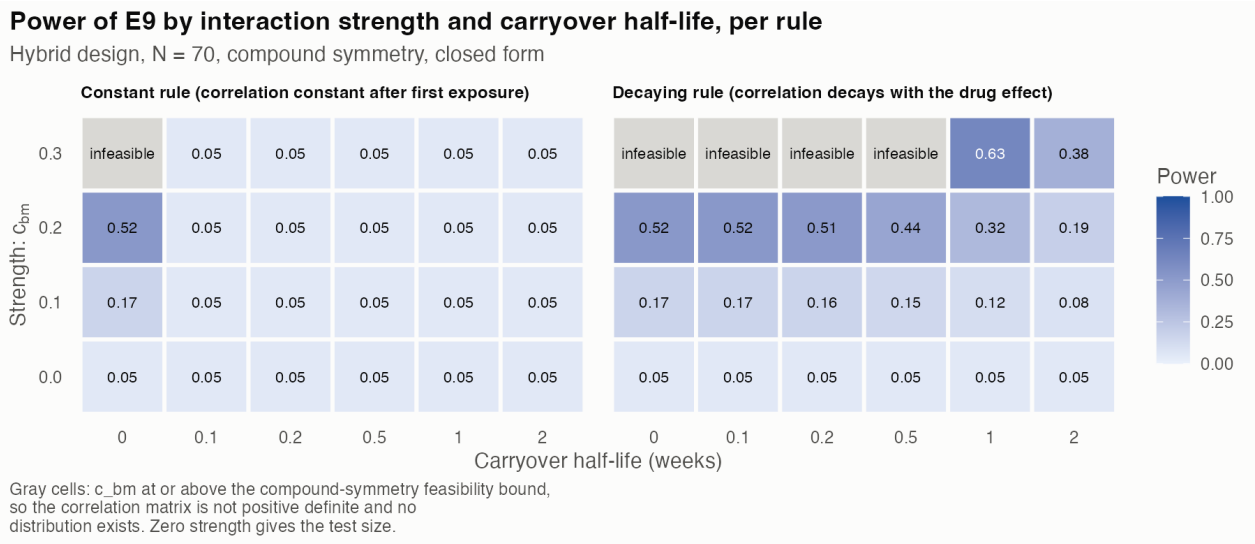


Figure 4: Closed-form power of the paired-difference statistic E9 by interaction strength (c_{bm}) and carryover half-life: the constant rule and the decaying rule. Gray cells: c_{bm} at or above the compound-symmetry feasibility bound, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry.

a biomarker that predicts the size of the drug effect cannot predict an effect that is no longer present. Two limitations of the choice are stated here. First, at the pharmacological half-lives of 0.1 to 0.2 week the on/off correlation difference is within 0.01 of 1. The decaying rule therefore returns the no-carryover result. The recommended DGP at its default half-life is not a test bed for carryover. The smooth decline is a statement about half-lives of 0.5 week and above, which we label behavioral. Second, the decaying rule is asymmetric. Off drug the moderation fades with the drug effect. On drug it is the full $c_{bm}\sigma_{BR}b_i$ at every visit, including the first weeks of titration when the Gompertz mean is a small fraction of its maximum. A correlation proportional to the current drug-response mean at every visit, or a correlation half-life separate from the mean's, would remove the asymmetry. Neither was tried. The sensitivity model of the Discussion is the form that addresses it. A companion paper takes up the mixed-model power of the recommended DGP at half-lives of 0.5 to 2 weeks (50). There the analysis half-life is either fixed or chosen from the data.

Stage 3. Carryover on the mean

The compounding carryover rule decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. Consider two off-drug visits at weeks 11 and 12 following discontinuation at week 10.

The compounding rule decays the week-10 value by $2^{-1/t_{1/2}}$ and then by $2^{-2/t_{1/2}}$, a total exponent of 3 weeks for 2 weeks elapsed. We treat the two forms as two specifications.

The anchored rule decays the mean from the last on-drug value, $\mu_{BR,\text{last}}\, 2^{-t_{sd}/t_{1/2}}$. The mean at any off-drug visit then depends only on the elapsed time, and the half-life means what it says (Appendix B.3).

The choice changes no slope, variance or feasibility bound. It changes only the path means. It therefore changes the closed-form power of study 1 by at most 0.002 in the Hybrid design. With the binary-indicator analysis it changes simulated power by at most 0.006. Figure 5 shows the drug-response mean by visit for the four Hybrid paths under both forms. Two features are visible. At $t_{1/2} = 1$ week the compounding rule lies below the anchored rule at the second off-drug visit, as expected. At 0.2 week both are indistinguishable from zero one week after discontinuation. At re-exposure (week 16 in paths 1 and 3) time on drug restarts at zero. The mean then rises along a fresh Gompertz curve to 4.3 points at the week-16 visit, four weeks into the restarted curve. The residual from the earlier exposure has decayed to 0.16 points by then even at a 1-week half-life. No downward step therefore appears on the Hybrid schedule. A step would appear only if a visit fell within about a week of re-exposure at a half-life of a week or more.

The recommended DGP decays carryover from the last on-drug mean. That choice rests on requirement 2: under the anchored rule the carryover half-life means the same thing for the mean as for the correlation, whereas the compounding rule decays faster than its stated half-life. Time on drug restarts at zero at re-exposure, which for prazosin matches re-titration after an interruption. That choice rests on pharmacological plausibility, as a judgment. For drugs without titration the response should instead rebuild from its residual level. That is an option, not the default. The numerical consequence of either choice is at most 0.006 in power.

Stage 4. Serial (AR(1)) correlation

Under compound symmetry, visits one week apart and sixteen weeks apart have the same correlation, ρ_{cs}. The one PTSD symptom series available to us, the total CAPS score in CSP 563 (planning section), has a correlation that decays slowly with lag. Matching its 4- and 20-week correlations within this DGP's structure would need ρ of about 0.87 to 0.91 per week (Additional file 1). That series is a total score, so it constrains the component-level ρ without determining it. Both compound symmetry and AR(1) are used in N-of-1 simulation (18). Serial correlation has been discussed as a

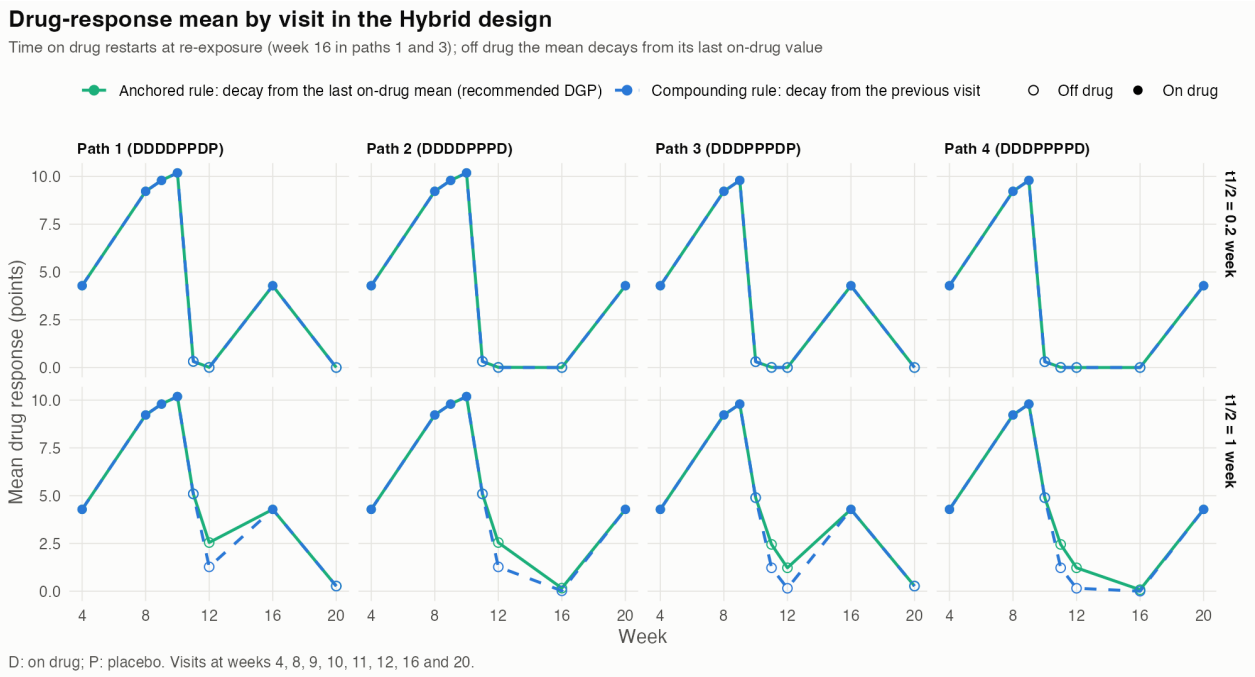


Figure 5: Drug-response mean by visit in the Hybrid design, by path, under the anchored carryover rule (decay from the last on-drug mean, the recommended DGP) and the compounding carryover rule (compounded decay), at carryover half-lives of 0.2 and 1 week. Filled points: on-drug visits; open points: off-drug visits. D: drug; P: placebo.

design consideration for personalized studies (53). Two elements are open: the form of the within-component correlation, and its value.

The recommended DGP uses AR(1) in calendar weeks, $\rho^{|w_t - w_s|}$, with one ρ for all three components. The value of ρ is a parameter. We take $\rho = 0.7$ as a convention only, not an estimate. Under the adopted separable form an effect size of 0.3 is already feasible at $\rho = 0.8$ (bound 0.440), so feasibility at 0.3 does not determine the value. Of the four values in Table 4, 0.7 has the largest separable feasibility bound (0.474). That keeps the planning effect sizes feasible but says nothing about whether 0.7 is right, and the value lies below the range the CSP 563 series suggests. Above 0.7, a larger ρ lowers the feasibility bound and raises E9 power (Table 4, 0.7 against 0.8). If that continues into the range CSP 563 suggests, the planning results at $\rho = 0.7$ overstate the largest feasible effect and the sample size needed. Table 5 extends the separable form to ρ from 0.4 to 0.95. The bound is largest at $\rho = 0.7$. It falls slowly below that value, to 0.437 at 0.4, and steeply above it, to 0.340 at 0.9 and 0.247 at 0.95, where the Hybrid design is the binding one. Above 0.7 the E9 sample size in the Hybrid design falls as well. The planning section repeats its calculations at $\rho = 0.5$ and 0.9.

Table 4 shows what ρ does under compound symmetry and under the structure adopted in Stage

Table 4: Serial correlation: the feasibility bound on c_{bm} (Ceiling; decaying rule, no carryover, minimum over the Hybrid, crossover and OL+BDC designs) and closed-form E9 power (Hybrid, $N = 70$, no carryover) by ρ, under compound symmetry and under the separable AR(1) form of Stage 5. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the separable form the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho \leq 0.7$, and the crossover design is never binding. At $t_{1/2} = 0.2$ week every power value is lower by 0.004 to 0.009.

	Compound symmetry		Separable AR(1)	
ρ_{cs} or ρ	Ceiling	Power at $c_{bm} = 0.25$	Ceiling	Power at $c_{bm} = 0.25$
0.5	0.458	0.492	0.448	0.261
0.6	0.410	0.553	0.462	0.270
0.7	0.346	0.628	0.474	0.290
0.8	0.256	0.721	0.440	0.335

5, for $\rho \in \{0.5, 0.6, 0.7, 0.8\}$. Under compound symmetry alone, lowering ρ_{cs} from 0.8 to 0.7 raises the feasibility bound from 0.256 to 0.346. An effect size of 0.3 then becomes feasible by the change of ρ alone, with no change of structure. The same change lowers closed-form power at $c_{bm} = 0.25$ from 0.721 to 0.628. A lower serial correlation leaves more within-person noise in the on-minus-off contrast. Feasibility and power pull in opposite directions. The feasible range widens as ρ falls, and the power at a matched c_{bm} falls.

The recommended DGP uses AR(1) in weeks at $\rho = 0.7$. The choice rests on requirement 1, the largest feasibility bound, traded against power. The value is uncalibrated. The AR(1) form is adopted for the reason given in Stage 5 (it is what the analysis model assumes), not because compound symmetry has been shown wrong for these data. Any study using the DGP should report ρ. Where the conclusion could depend on it, the analysis should be repeated at other values, as the planning section does at 0.5 and 0.9.

Stage 5. The separable cross-component structure

Under compound symmetry the cross-component correlation is $c_1 = 0.2$ at the same visit and $c_\times = 0.1$ at every other lag. With AR(1) within components, the direct translation keeps c_1 at lag zero and lets the other-visit correlation decay, $c_\times \rho^{|w_t - w_s|}$. This two-constant form is what the package's `buildSigma()` builds when its two constants differ (Appendix B.5). It is not the only choice, and it has a validity condition.

The recommended DGP uses the separable form, $c_1 \rho^{|w_t - w_s|}$ at every lag including zero. The response

Table 5: Serial correlation under the separable AR(1) form of the recommended DGP ($c_1 = 0.2$, decaying rule, carryover decaying from the last on-drug mean): the feasibility bound on c_{bm}, minimum over the Hybrid, crossover and OL+BDC designs, by carryover half-life, the design that attains the minimum without carryover, and the closed-form total N for 80 percent power of E9 in the Hybrid design at the 0.2-week half-life. Not feasible: c_{bm} lies above the Hybrid bound.

	Feasibility bound				Hybrid N, E9	
ρ	$t_{1/2} = 0$	0.2 week	1 week	Binding design	$c_{bm} = 0.25$	$c_{bm} = 0.35$
0.40	0.437	0.438	0.445	OL+BDC	318	160
0.50	0.448	0.450	0.465	OL+BDC	314	158
0.60	0.462	0.465	0.494	OL+BDC	302	152
0.65	0.469	0.473	0.512	OL+BDC	292	148
0.70	0.474	0.479	0.533	OL+BDC	278	140
0.75	0.465	0.471	0.550	Hybrid	258	130
0.80	0.440	0.446	0.544	Hybrid	234	118
0.85	0.400	0.407	0.519	Hybrid	200	100
0.90	0.340	0.346	0.457	Hybrid	154	not feasible
0.95	0.247	0.251	0.343	Hybrid	92	not feasible

block is then a Kronecker product $M = K \otimes A$ of a 3×3 component correlation K (1 on the diagonal, c_1 off it) and one AR(1) kernel $A_{ts} = \rho^{|w_t - w_s|}$. In words, every component has the same first-order autoregressive correlation, and the between-component correlation is constant at every lag. It is the two-constant form with $c_\times = c_1$, with the default $c_1 = 0.2$. Four properties follow (Appendix A.5).

1. **Validity for every schedule.** The eigenvalues of M are products of those of K and A, so M is positive definite for any visit times. The two-constant form adds a same-visit cross-component term that carries correlation without variance. It is positive definite only when the smallest eigenvalue of A exceeds $(c_1 - c_\times)/(1 - c_\times) = 0.111$. On the paper's schedules the condition holds (smallest eigenvalue 0.122 in the Hybrid and OL+BDC designs at $\rho = 0.8$, 0.193 at 0.7; 0.28 and 0.43 in the crossover design). On 20 weekly visits at $\rho = 0.8$ it holds by less than 0.001. On daily measurement over eight weeks it fails (0.016 at $\rho = 0.8$, 0.026 at 0.7). The two-constant form therefore cannot be used for the dense schedules that wearable or diary outcomes produce.
2. **The compound-symmetry structure is already of this kind in pieces.** It is the sum of a person-level and an occasion-level separable term, each valid. This is why its matrices fail only through the biomarker row, never in the response block.
3. **An exact, interpretable feasibility bound.** The bound factors into a component term

and a schedule term,

$$c^*_{bm} = \sqrt{1 - R^2}\,\left(g^\top A^{-1} g\right)^{-1/2}, \qquad R^2 = \frac{2c_1^2}{1 + c_1},$$

where R^2 is the squared multiple correlation of the drug response on the other two components at one visit. The schedule term is a sum of per-step terms of the correlation profile. A profile that switches abruptly between closely spaced visits has a large per-step term. This is why designs with many on-off switches can carry only small effects.

4. **It nearly matches the analysis model.** Under separability the summed response has covariance $(s_t^\top K s_s)\,\rho^{|w_t - w_s|}$, with s_t the component SDs at visit t (Appendix A.5). With constant SDs this is exactly AR(1), the residual structure `corCAR1` assumes. In the Hybrid design the placebo-response SD is $10\eta_t$, so it halves at blinding. The summed-response variance is then 368 at open-label visits and 257 at blinded visits. The correlation between an open-label and a blinded visit is the AR(1) value times 0.976. The summed response is therefore AR(1) only within a phase. Under the two-constant form it is, in addition, AR(1) plus a white-noise term the analysis model omits. These statements follow from the algebra. Their effect on test calibration has not been measured separately.

Table 6 gives the feasibility bounds. At $\rho = 0.8$ the two-constant form raises the bound only to 0.291. A single same-visit correlation repeated across visits that share little demands more of the biomarker row. In the open-label design, where every visit is on drug, it lowers the bound. The separable form raises the bound to 0.440. At $\rho = 0.7$ the two forms are close, 0.452 against 0.474. At the recommended ρ the separable form's gain in feasible range is therefore small, and its reasons are properties 1 and 4. The bound of 0.474 is attained in the OL+BDC design (0.480 in the Hybrid design). A correlation of 0.45 is feasible in every design without carryover. Effects as large as 0.6 exceed every feasibility bound in the Hybrid design.

The wider feasible range has a consequence that does not depend on the form. At the same c_{bm}, AR(1) data carry less information about the interaction than compound-symmetric data. Take the closed-form power of the paired-difference statistic in the Hybrid design at $c_{bm} = 0.25$. It is 0.721 under compound symmetry, 0.335 under the separable form at $\rho = 0.8$ and 0.290 at $\rho = 0.7$. A larger feasible correlation does not by itself mean more power. Effect sizes under different structures should therefore be compared on the scale of the implied slope, not of c_{bm}.

Table 6: Largest feasible biomarker-response correlation c_{bm} under the decaying rule, without carryover, minimum over the Hybrid, crossover and OL+BDC designs. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the two AR(1) forms the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho = 0.7$ (separable form: Hybrid 0.480), and the crossover design is never binding.

ρ_{cs} or ρ	Compound symmetry	AR(1), two-constant cross-component	Separable AR(1)
0.8	0.256	0.291	0.440
0.7	0.346	0.452	0.474

The recommended DGP uses the separable form with $c_1 = 0.2$. The choice rests on requirement 1: the form is positive definite for any schedule and gives an exact feasibility bound (properties 1 and 3). Its near agreement with the analysis model's residual structure (property 4) is a further advantage. The value $c_1 = 0.2$ is the default value and is uncalibrated.

Stage 6. Optional features

The recommended DGP has three optional features: phase-specific measurement error, a Weibull decay form with the same half-life, and monotone dropout. Study 2 fitted ten reported analyses to trials with and without each feature. This stage reports what the features did to size and power. The biomarker is not prognostic in any cell reported here (Aims). The study's cells with a prognostic biomarker are reported in the companion paper (45).

Size. We pooled the 11 null cells (11,000 replicates). The nominal band is 0.05 plus or minus two Monte Carlo SE, $2\sqrt{0.05 \times 0.95/11{,}000} = 0.004$, that is 0.046 to 0.054. The binary-indicator analysis rejected 0.069, and 0.077 at $N = 35$; both use the working-covariance standard error. The unstructured model, with the working-covariance standard error and Kenward-Roger inference (54), rejected 0.055, with 0.084 at $N = 35$. Random-intercept models with continuous-time AR(1) residuals and the cluster-robust standard error (CR2) (47,48), at the fixed analysis half-life, rejected 0.048 to 0.049. No single cell of these models exceeded 0.067. With the half-life chosen by AIC, the CAR(1) model rejected 0.053 and the expectancy-term model 0.056. The latter's largest single cell was 0.071, under $t_{1/2} = 1$ week. Of the analyses in this paragraph, the binary-indicator analysis, the unstructured model and the expectancy-term model with AIC-chosen half-life lie above the pooled band. Single-cell flags should be judged against the number of comparisons. With 10 reported analyses and 11 null cells, about 5 analysis-cell pairs would fall outside a two-MCSE band by chance at nominal size. The pairs are not independent, since all analyses share the same trials. A

Table 7: The optional features of the recommended DGP: null rejection rate ($c_{bm} = 0$; 1,000 replicates, MCSE 0.007) and power ($c_{bm} = 0.30$; 500 replicates, MCSE at most 0.022) for the binary-indicator analysis with the working-covariance standard error (Binary), and the random-intercept CAR(1) model and the expectancy-term model (Expect.), both with the cluster-robust standard error (CR2). Hybrid design, third run of study 2 (the recommended DGP: carryover decaying from the last on-drug mean, fixed allocation). Each row changes one feature of the base cell.

	Null rejection			Power		
Cell	Binary	CAR(1)	Expect.	Binary	CAR(1)	Expect.
Base cell ($N = 70$, $t_{1/2} = 0.5$, exponential, no dropout)	0.077	0.051	0.067	0.474	0.486	0.404
Measurement error (SD 3 and 6)	0.069	0.050	0.037	0.422	0.394	0.314
Dropout 20 percent, completely at random	0.078	0.053	0.055	0.466	0.464	0.392
Dropout 20 percent, dependent on last score	0.069	0.060	0.047	0.424	0.426	0.360
Weibull decay, shape 0.5	0.072	0.046	0.044	0.378	0.400	0.300
Weibull decay, shape 2	0.066	0.039	0.046	0.526	0.540	0.496
$t_{1/2} = 0$	0.071	0.040	0.041	0.536	0.566	0.498
$t_{1/2} = 1$ week	0.066	0.047	0.061	0.418	0.416	0.348
$N = 35$	0.077	0.053	0.048	0.270	0.246	0.206
$N = 150$	0.060	0.048	0.045	0.768	0.790	0.688

single exceedance is therefore weak evidence on its own.

The options. Table 7 shows what each optional feature did to size and power. It covers three analyses: the binary-indicator analysis with the working-covariance standard error, the random-intercept CAR(1) model with the cluster-robust standard error, and the expectancy-term model with the cluster-robust standard error, which adds the design expectancy and a biomarker-by-expectancy term to the mean model. Measurement error (SD 3 open label, 6 blinded) lowered power by 0.05 to 0.09. Dropout of 20% completely at random lowered power by 0.01 to 0.02. Dropout of 20% dependent on the last score lowered power by 0.04 to 0.06. A Weibull decay with the same half-life lowered power by 0.09 to 0.10 at shape 0.5 (a steep early decay) and raised power by 0.05 to 0.09 at shape 2. The true half-life itself moved power from 0.50 to 0.57 at zero to 0.35 to 0.42 at one week. Halving N gave power of 0.21 to 0.27, against 0.40 to 0.49 in the base cell. At $N = 150$ the CAR(1) model reached 0.79. In these null cells the CAR(1) and expectancy-term models rejected at most 0.067, against a single-cell band of 0.036 to 0.064 (the maximum, for the expectancy-term model under the base cell, lies 0.003 above the upper edge). The binary-indicator analysis rejected up to 0.078, under dropout completely at random. The features change power, which the planning section and the companion paper address.

The recommended DGP includes the three features as options whose effect on size and power is tabulated. The choice rests on requirement 3, flexibility for stress conditions. Measurement error is implemented in the stress-test driver and not yet in the package (Appendix B.5). The package also implements prognostic pathways, off by default; they are outside the scope of this paper (45).

Stage 7. The recommended DGP

Table 8 collects the choices. Each row is marked as a choice made in a stage, a default value, an option, or an assumption, and names the section that justifies it.

Table 8: The recommended data-generating process. The Default column gives the value of each element in the recommended DGP; where the stress test of Stage 6 used another value it is stated. The stress test's reported run uses fixed allocation; its second run assigned participants to paths at random, a driver option outside the recommended DGP (Appendix B.5).

Element	Specification	Default	Status	Justified in
Components	$Y_{it} =$ $\mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, modified Gompertz means	default values (Methods)	default	Methods
Feasibility	stop if the correlation matrix is not positive definite	always	choice	Stage 1
Decaying biomarker-response correlation	c_{bm} on drug; $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure	c_{bm} below the feasibility bound of every design in the study	choice	Stage 2
Carryover on the mean (anchored)	decay from the last on-drug value, $\mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$	$t_{1/2} = 0.1$ to 0.2 week (pharmacological, assumed); 0.5 in the stress test	choice	Stage 3
Re-exposure	time on drug restarts at zero	on, for titrated drugs	choice	Stage 3

Element	Specification	Default	Status	Justified in
Serial (AR(1)) correlation	$\rho^{\lvert w_t - w_s \rvert}$ in calendar weeks, one ρ for all three components	$\rho = 0.7$ (convention; report and vary)	choice	Stage 4
Separable cross-component structure	$c_1\rho^{\lvert w_t - w_s \rvert}$ at every lag, applied to the correlation, since the PB SD scales with the expectancy	$c_1 = 0.2$ (default value)	choice	Stage 5
Path allocation	fixed numbers	18, 18, 17, 17 at $N = 70$ (Hybrid)	default	Methods
Prognostic association	none: the biomarker is correlated with BR only, not with TV, PB or baseline severity	zero (the prognostic case is treated in the companion paper)	assumption	Aims
Measurement error	independent, SD by phase	off; 3 open label, 6 blinded in the stress test	option (driver only)	Stage 6
Decay form	exponential or Weibull with the same half-life	exponential	option	Stage 6
Dropout	monotone, completely at random or at random given the last score	off; 20% in the stress test	option	Stage 6

Using the recommended DGP to plan a trial

What c_{bm} means. Conditional on the biomarker, the correlation encoding shifts the on-drug drug-response mean by $c_{bm}\sigma_{BR}$ per standard deviation of biomarker (Discussion). With the default $\sigma_{BR} = 8$, $c_{bm} = 0.25$ is a slope of 2.0 symptom points per SD of biomarker, 0.35 is 2.8 points, 0.45 is 3.6 points, and the feasibility bound of 0.474 is 3.8 points. The biomarker's default SD is 15.4 mm Hg, so 0.25 is 0.13 points per mm Hg. The association first reported for prazosin is 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg (11), or 21.5 points per SD of biomarker.

Representing that slope under the correlation encoding would need $\sigma_{BR} \geq 45$ at the bound. That is larger than the SD of the symptom score itself (18.5 at baseline at the default values). At the default values the bound is therefore below the reported association. The comparison is approximate for two reasons. The observed slope is on the total symptom change, not the drug-response component. And $\sigma_{BR} = 8$ is a default value, not an estimate.

That association did not replicate in our reanalysis of CSP 563 (Additional file 1). The interaction was +0.25 points per 10 mm Hg (95% CI −2.72 to 3.21). With $\sigma_{BR} = 8$ and the trial's biomarker SD of 14.2 mm Hg, this interval corresponds to c_{bm} from −0.57 to 0.48. The values of c_{bm} used in this paper, 0.25 to 0.45, imply 1.4 to 2.5 points per 10 mm Hg. They lie inside that interval. CSP 563 therefore neither supports nor excludes them, and it gives no reason to plan for a slope above the feasibility bound. A study that does need a larger slope should use the sensitivity model of the Discussion once it is implemented, or raise σ_{BR} with a stated reason.

What CSP 563 identifies. The reanalysis estimates some quantities of the DGP on the total symptom score, all as computed in Additional file 1 and not re-estimated here. The baseline SD is 16.0 (DGP: 18.5) and the biomarker SD 14.2 mm Hg (DGP: 15.4). The biomarker is not prognostic for baseline severity ($r = 0.02$, 95% CI −0.09 to 0.13), which supports the assumption of a zero prognostic correlation with baseline (Aims). The placebo-arm improvement has the DGP's time course and is about 1.8 times its size. The SD of change from baseline grows from 17 to 24 points over 26 weeks, where the DGP holds it at 16.0. The total-score correlation decays by about 0.975 per week with about 10 percent measurement noise. The DGP's total-score correlation falls faster at short lags and then flattens. The trial does not identify the split between the natural-history and placebo-response components, the expectancy weight, the component-level correlations or the carryover half-life. All participants were blinded throughout, and no one stopped drug. A sensitivity arm of the DGP calibrated to these estimates is future work. A working protocol for it exists in the repository, and it has not been run.

Sample size. Table 9 gives the total N for 80% power under the recommended DGP, by design, c_{bm} and half-life: in closed form for the paired-difference statistic E9, and by simulation for the recommended analysis G10 in the Hybrid design. At the pharmacological half-life the Hybrid design needs about 280 participants at $c_{bm} = 0.25$ and 84 at 0.45. At a 1-week half-life those rise to 480 and 146.

The recommended analysis at the planned sample sizes. The companion paper (50) recom-

mends the analysis G10 for the interaction. G10 is a linear mixed model with a random intercept and continuous-time AR(1) residual correlation. Its drug indicator is the exposure-decayed D_{bc} at an analysis half-life chosen by AIC over 0.25, 0.5, 1 and 2 weeks. The selection uses maximum-likelihood fits of the model without the interaction, so the choice of half-life does not depend on the term under test. The model at the chosen half-life is refitted by REML with the interaction, and the interaction is tested with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom. The grid of half-lives does not include 0 or the pharmacological values of 0.1 to 0.2 week, so at those true values G10 uses an analysis half-life of at least 0.25 week. The AIC-chosen analyses of the stress test also consider 0 (Table 1). We simulated G10 under the recommended DGP with the default values in every cell of Table 9, at $N = 70$ and at the E9 sample size for 80 percent power, with 1,000 replicates per cell. We also simulated the null ($c_{bm} = 0$) at $N = 70$ for each design and half-life with 2,000 replicates. Table 10 gives the rejection rates at the 0.2- and 1-week half-lives. Where the E9 sample size exceeded 400, or was above 800 and not computed, the cell ran at $N = 400$ and is marked.

Three results follow. First, G10 rejected the null at 0.046 to 0.053 across the nine design-by-half-life cells, including the half-life of 0 not shown in the table (Monte Carlo SE at most 0.005). Second, at $N = 70$ G10 has power more than two Monte Carlo SEs above that of E9 in 23 of the 27 cells, within two SEs in 4 and below in none. In the Hybrid design at $c_{bm} = 0.25$ and the 0.2-week half-life, G10 power is 0.357 (MCSE 0.015) against 0.285 for E9. In the crossover design at the same cell it is 0.243 (MCSE 0.014) against 0.202. Third, at the E9 sample size for 80 percent power, G10 power is 0.844 to 0.936 in the 20 uncapped cells, and 0.601 to 0.915 in the 7 cells that ran at $N = 400$. Under the recommended DGP at $\rho = 0.7$ with the default values, the closed-form E9 rule is therefore conservative for G10. A trial sized by Table 9 and analyzed with G10 had power of at least 0.844 in every uncapped cell, with the smallest margin in the crossover design. Fits failed to converge in at most 0.008 of replicates in any cell, and the rejection rates are computed over converged fits.

The same simulated trials were also analyzed with G9, which chooses the half-life by AIC with the interaction in the model. In every cell the G9 and G10 rejection rates differ by at most 0.040.

The sample size for G10. In the nine Hybrid cells we simulated G10 at 6 multiples of the E9 sample size, from 0.35 to 1.25 times, with 500 replicates at each. In each cell we fitted a probit regression of rejection on $\log N$. The G10 columns of Table 9 give the N at which the fitted power

is 0.80. That N lies inside the simulated range in all 9 cells, and it is 0.71 to 0.83 times the E9 value. The E9 values are closed form, so this comparison is the measured difference between the paired-difference approximation and the analysis a trial would use.

Serial correlation. The planning results above assume $\rho = 0.7$, a convention (Stage 4). We repeated the closed-form calculation and the G10 simulation at $\rho = 0.5$ and 0.9, with the same seeds and replicate counts. At 0.5 the grid of c_{bm} is unchanged, and one cell lies above its bound (OL+BDC at 0.45 without carryover). At 0.9 the Hybrid bound falls to 0.340, so the grid moves to 0.20, 0.25 and 0.30. Table 11 gives the cells at the 0.2-week half-life. Below, ranges over all half-lives cover 0, 0.2 and 1 week.

In the Hybrid design the required sample size depends strongly on ρ above 0.7 and little below it. At $c_{bm} = 0.25$ the E9 rule gives 314, 278 and 154 participants at $\rho = 0.5$, 0.7 and 0.9. The crossover needs 1.5 times the Hybrid sample at 0.7 and 2.6 times at 0.9. The likely reason is visit spacing. The Hybrid design compares on- and off-drug visits one week apart, where a high serial correlation cancels most of the shared noise. The crossover compares two blocks of four visits 2.5 weeks apart, ten weeks apart on average. G10 rejected the null at 0.047 to 0.056 in the 18 design-by-half-life cells at the two new values. At the E9 sample size its power was 0.815 to 0.867 at $\rho = 0.5$ and 0.938 to 0.999 at 0.9, in the uncapped cells. The E9 rule therefore stays conservative for G10, and at $\rho = 0.9$ it is very conservative. There G10 had power 0.751 in the Hybrid design at $c_{bm} = 0.25$ with only 70 participants. We repeated the G10 sample-size search in the Hybrid design (Table 11). Over all three half-lives it gives 0.80 to 0.91 times the E9 value at $\rho = 0.5$ and 0.45 to 0.57 times at 0.9, against 0.71 to 0.83 at 0.7. At both new values the sample size lies inside the simulated range in every cell.

In the crossover design at $\rho = 0.9$, 8 to 12 percent of G10 fits failed in each cell at $N = 70$, and 13 to 19 percent at the E9 sample sizes below 400. In the cells that ran at $N = 400$ at most 0.1 percent failed. In every other cell at the two new values at most 2.6 percent failed. In the failed replicates we refitted, the model chosen by AIC put the serial correlation at its upper limit of 1 and the random-intercept SD near zero. A correlation that does not decay cannot be told apart from a random intercept, and the CR2 step then failed because the fitted covariance was singular. The rates in Table 11 are over the fits that succeeded. A trialist who expects slow decay and plans a crossover analyzed with G10 should expect such failures and specify a fallback model in advance.

When these sample sizes apply. The sample sizes assume, as the whole paper does, that the

Table 9: Total N for 80 percent power at $\alpha = 0.05$ under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying biomarker-response correlation, carryover decaying from the last on-drug mean, default values), by design, interaction strength and carryover half-life, with the implied slope in symptom points per SD of biomarker ($\sigma_{BR} = 8$). E9: closed form. G10: Hybrid design only; the N at which a probit regression of rejection on $\log N$, fitted to simulations at 6 multiples of the E9 value (500 replicates each), reaches 0.80, rounded up. Without carryover the E9 values are at most 3 percent below those at 0.2 week. Participants are allocated to paths in fixed proportions (18, 18, 17, 17 in the Hybrid design at $N = 70$).

			E9 (closed form)		G10 (simulated)	
Design	c_{bm}	Slope (points per SD)	0.2 week	1 week	0.2 week	1 week
Hybrid	0.25	2.0	278	480	205	384
	0.35	2.8	140	244	108	193
	0.45	3.6	84	146	64	121
OL+BDC	0.25	2.0	404	> 800		
	0.35	2.8	206	452		
	0.45	3.6	124	272		
Crossover	0.25	2.0	424	454		
	0.35	2.8	216	230		
	0.45	3.6	130	138		

biomarker is not prognostic (Aims). G10 does not hold its level under every prognostic pathway. A trial in which such a pathway is plausible should be planned for an analysis that does, and its sample size is larger; the companion paper gives both (45).

Parameters by status.

- *Default values:* the Gompertz parameters of the three components and their standard deviations; the biomarker and baseline means and standard deviations; $c_1 = 0.2$; the path allocation; the expectancy values.
- *Chosen here:* the restart of time on drug at re-exposure; the decaying biomarker-response correlation and the stop on infeasible matrices; carryover decaying from the last on-drug mean; AR(1) in weeks with $\rho = 0.7$; the separable cross-component form; the symptom half-life range of 0.1 to 0.2 week; the option defaults used in the stress test.
- *Uncalibrated:* every value above. None was re-estimated for this paper. CSP 563 estimates the baseline and biomarker SDs, the total placebo-arm improvement and the total-score covariance, and it does not identify the component split, the expectancy weight or the half-life (above). The values to which the results are most sensitive are ρ (Stage 4), σ_{BR} (which sets the slope a given c_{bm} implies) and the half-life.

Table 10: Rejection rate of the analysis G10 under the recommended DGP of Table 9, by design, interaction strength and carryover half-life, at $N = 70$ and at the closed-form E9 N for 80 percent power. Rows with $c_{bm} = 0$ give the size at $N = 70$ (2,000 replicates per cell, Monte Carlo SE at most 0.005); the other rows give power (1,000 replicates per cell, Monte Carlo SE at most 0.016). †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Rejection rates are over converged fits (at most 0.8 percent of replicates failed in any cell).

		At $N = 70$		At the E9 N	
Design	c_{bm}	0.2 week	1 week	0.2 week	1 week
Hybrid	0.00	0.051	0.048		
	0.25	0.357	0.217	0.900	0.809†
	0.35	0.610	0.362	0.899	0.885
	0.45	0.836	0.531	0.903	0.874
OL+BDC	0.00	0.051	0.051		
	0.25	0.275	0.133	0.915†	0.601†
	0.35	0.507	0.235	0.936	0.866†
	0.45	0.722	0.362	0.932	0.891
Crossover	0.00	0.046	0.050		
	0.25	0.243	0.220	0.840†	0.812†
	0.35	0.365	0.378	0.867	0.862
	0.45	0.592	0.551	0.869	0.844

Table 11: Planning by serial correlation at the 0.2-week half-life, as Table 9 (which gives $\rho = 0.7$) but at $\rho = 0.5$ and 0.9: the closed-form total N for 80 percent power of E9, the G10 N for 80 percent power (Hybrid design only), G10 power at the E9 N (1,000 replicates per cell), and the percent of G10 fits that failed at $N = 70$. At $\rho = 0.9$ the grid of c_{bm} lies inside the lower bound. Not feasible: c_{bm} lies above the bound of that design. †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Power is over the fits that succeeded.

ρ	Design	c_{bm}	E9 N	G10 N	G10 power at the E9 N	G10 failed (%)
0.5	Hybrid	0.25	314	268	0.849	0
		0.35	158	140	0.829	0
		0.45	96	86	0.822	0
	OL+BDC	0.25	412		0.844†	0
		0.35	208		0.859	0
		0.45	126		0.856	0
	Crossover	0.25	348		0.864	0
		0.35	176		0.825	0
		0.45	106		0.815	0
0.9	Hybrid	0.20	242	117	0.979	0
		0.25	154	74	0.976	0
		0.30	106	53	0.979	0
	OL+BDC	0.20	462		0.992†	0
		0.25	296		0.999	0
		0.30	204		0.997	0
	Crossover	0.20	628		0.803†	10
		0.25	402		0.944†	8
		0.30	278		0.938	9

Choosing c_{bm}. Take the smallest feasibility bound over the designs and half-lives in the study (Table 6 and Appendix A.5; 0.474 for the three designs at $\rho = 0.7$ without carryover, higher with carryover). State c_{bm} as a slope in outcome points per SD of biomarker. Keep it below that bound. The package reports an infeasible value by stopping.

Differences from the DGP of Hendrickson et al.

The DGP of this paper builds on Hendrickson et al. (2), whose code at commit 58b32a9 we take as their reference. Table 12 lists every difference and names the section that justifies it. The rest of this section reports what the differences do when they are applied to their code.

Table 12: Differences between the DGP of Hendrickson et al. (code at commit 58b32a9) and the recommended DGP of this paper.

Element	Hendrickson et al.	This paper	Justified in
Infeasible matrices	replaced by a nearby valid matrix (Higham's nearest positive semidefinite matrix, tolerance 10^{-3}), without a warning	refused: the cell stops and is reported	Stage 1; Appendix B.2
Correlation off drug	constant: c_{bm} wherever the drug-response mean is nonzero, so the full value at every visit after first exposure	decaying: $c_{bm}2^{-t_{sd}/t_{1/2}}$ after exposure, 0 before	Stage 2; Appendix B.1
Mean off drug	compounding: each off-drug mean decayed from the previous visit's	anchored: decayed from the last on-drug mean	Stage 3; Appendix B.3
Serial correlation	compound symmetry, $\rho_{cs} = 0.8$	AR(1) in calendar weeks, $\rho = 0.7$	Stage 4
Cross-component correlation	$c_1 = 0.2$ at the same visit, $c_\times = 0.1$ at every other lag	separable, $c_1\rho^{\lvert w_t - w_s\rvert}$ at every lag, $c_1 = 0.2$	Stage 5
Dropout weight	weight rises with improvement (100 plus the improvement)	weight rises with worsening (100 minus the improvement, floored at zero)	Appendix B.5
Optional features	none	measurement error, Weibull decay, dropout	Stage 6

Unchanged from their DGP: the component decomposition, the Gompertz means and clocks, the parameter values, the correlation encoding, the restart of time on drug at re-exposure, fixed path

allocation, and the absence of a prognostic association (Methods).

Their code, as published and with the decaying rule

We reran their code in two versions: as published, and with only the correlation lines replaced by the decaying rule (Appendix B.1). Both versions used their analysis, the binary-indicator mixed model fitted with `lmer` and tested with Satterthwaite degrees of freedom, and the same seeds. We used $c_{bm} = 0.25$, below the compound-symmetry feasibility bound of 0.256, so that no matrix would be repaired and the correlation rule would be the only difference. Every replicate returned an estimate and no matrix was repaired. The rerun had two parts: (a) the Hybrid design at $N = 70$ without censoring, at $t_{1/2} \in \{0, 0.1, 0.2, 1\}$, with 500 replicates per cell (Monte Carlo SE of power at most 0.022) and one seed per half-life; and (b) the four designs at $N \in \{35, 70\}$ and $t_{1/2} \in \{0, 0.1, 0.2\}$ under their censoring patterns, with 250 replicates per cell (at most 0.032) and one seed per design, sample size and half-life.

As published. Power fell from 0.668 without carryover to 0.126 at a half-life of 0.1 week (Table 13), the loss that Hendrickson et al. reported. At that half-life only 0.1% of the drug effect remains a week after stopping. This is the constant rule of Stage 2, under which no interaction remains among the post-baseline visits.

In our own implementation of their DGP, the mean estimate falls from -0.126 to between -0.026 and -0.032 at every positive half-life, and power from 0.63 to between 0.08 and 0.11 (500 replicates per cell). Their code gives 0.126 at 0.1 week against 0.080 for our implementation. The difference is 2.4 combined Monte Carlo SE (two-sided $p = 0.016$). The two implementations share the parameters, the path allocation and the analysis model. They differ in the random-number stream and in the carryover rule on the mean. At 0.1 week the carryover rule leaves an off-drug drug-response mean below 0.011 points under either rule, so it cannot account for a difference of this size. We therefore reran their code at 0.1 week with 4,000 new replicates (16 seeds not used before). Power was 0.090 (MCSE 0.0045), within 0.8 combined Monte Carlo SE of our implementation. Rerunning the stored cell with its stored seed reproduces 0.126 exactly. The value of 0.126 is therefore consistent with a high draw of a 500-replicate run. We found no difference between the two implementations that would explain it.

Why their analysis keeps a little power. Under the constant rule the paired-difference statistic E9 has power equal to the test size (Stage 2), but the binary-indicator analysis does not. It counts

Table 13: The code of Hendrickson et al. at commit 58b32a9, as published (constant) and with the decaying biomarker-response correlation (decaying): Hybrid design, $N = 70$, $c_{bm} = 0.25$, binary-indicator analysis, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022. A rerun of the published version at 0.1 week with 4,000 new replicates gave power 0.090.

$t_{1/2}$ (weeks)	Mean estimate, constant	Mean estimate, decaying	Power, constant	Power, decaying
0	-0.131	-0.131	0.668	0.668
0.1	-0.028	-0.129	0.126	0.638
0.2	-0.029	-0.131	0.092	0.656
1	-0.033	-0.107	0.094	0.462

baseline as an off-drug visit. The paired-difference statistic coded the same way, E9b, shows what that adds. Every post-baseline score is the baseline score minus the summed responses, so baseline cancels from the contrast. Counting it as off drug only shrinks the off-drug mean of the responses by $n_{\text{off}}/(n_{\text{off}} + 1)$ toward their baseline value of zero. At baseline the correlation is zero, so under the constant rule

$$\beta^{\text{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\frac{1}{n_{\text{off}} + 1} \qquad \text{for every } t_{1/2} > 0,$$

a quarter or a fifth of the no-carryover slope, with power 0.081 instead of 0.050. E9b's pooled slope, -0.029, matches the binary-indicator analysis's mean estimates of -0.026 to -0.032. This residual compares the on-drug visits with the pre-treatment visit. It is not an interaction during the trial.

With the decaying rule. Replacing only the correlation lines and running their pipeline unchanged gave the results in Table 13. The two versions coincide without carryover, as they must. At 0.1 and 0.2 week the decaying version keeps the full interaction, within Monte Carlo error of the no-carryover value of 0.668. At 1 week it shows a gradual decline, consistent with the monotone decline of Stage 2.

Figure 6 repeats this comparison in the layout of their Figure 4B: all four designs, both sample sizes and their five censoring patterns. It uses $c_{bm} = 0.25$ instead of their 0.6. The value 0.6 lies above the feasibility bound of the Hybrid design and of most cells of the other designs, so it would confound the comparison with the repair. With 250 replicates per cell the Monte Carlo SE reaches 0.032, so a difference of 0.05 between adjacent cells is within Monte Carlo error. Only the pattern across cells should be read. Results at $N = 70$ without censoring:

- **Hybrid design.** Power under the constant rule falls from 0.66 to 0.08 at both positive

half-lives. Power under the decaying rule stays at 0.62 and 0.68.

- **Open label plus blinded discontinuation.** Power under the constant rule falls from 0.57 to 0.12 and 0.13. Power under the decaying rule stays at 0.57 and 0.60.
- **Crossover.** Power under the constant rule falls only to 0.34 and 0.33. The placebo-first path's off-drug visits precede any exposure and so keep zero correlation under both rules. Power under the decaying rule stays at 0.71 and 0.78.
- **Open label.** There are no off-drug visits. The two rules coincide and power is low throughout.

The same pattern holds at $N = 35$ and under every censoring pattern. Their censoring patterns raise dropout with improvement, through a sign error in the biased term; the figure keeps them, because it reproduces their code.

Against the recommended DGP. G10 power under the recommended DGP (Figure 3) is lower than in their code with the decaying rule (Figure 6, lower two panel rows) in every design. Without carryover it is 0.34, 0.34 and 0.20 in the Hybrid, OL+BDC and crossover designs, against 0.66, 0.57 and 0.74 in their code. The analysis does not account for the difference. Under the recommended DGP the closed-form power of the paired-difference statistic E9 at the same N and c_{bm} is 0.29, 0.21 and 0.20, computed as for Table 9, no higher than G10. Without carryover the two carryover rules on the mean coincide, so the difference comes from the correlation structure: compound symmetry at $\rho_{cs} = 0.8$ in their code, the separable AR(1) form at $\rho = 0.7$ here. Table 4 shows the effect of that change on closed-form E9 power in the Hybrid design. The drop is largest in the crossover design. We have not established why.

Repaired matrices in their grid

The nonzero effect sizes of their grid were 0.3 and 0.6, in the open-label, open-label-plus-blinded-discontinuation, crossover and Hybrid designs at $t_{1/2} \in \{0, 0.1, 0.2\}$ week. Counting one matrix per design path, effect size and half-life, 20 of its 54 matrices at those effect sizes (37%) were replaced by nearby valid matrices. The counts from our calculation reproduce those of their code exactly. At $c_{bm} = 0.3$ the repair changed no correlation by more than 0.007. Results at that effect size are therefore essentially those of the stated DGP. At $c_{bm} = 0.6$ the repair lowered the on-drug correlation to 0.54 to 0.57 and moved other correlations by up to 0.10. Those results describe a different DGP. The constant rule passes more cells at positive half-lives, because a constant correlation places a

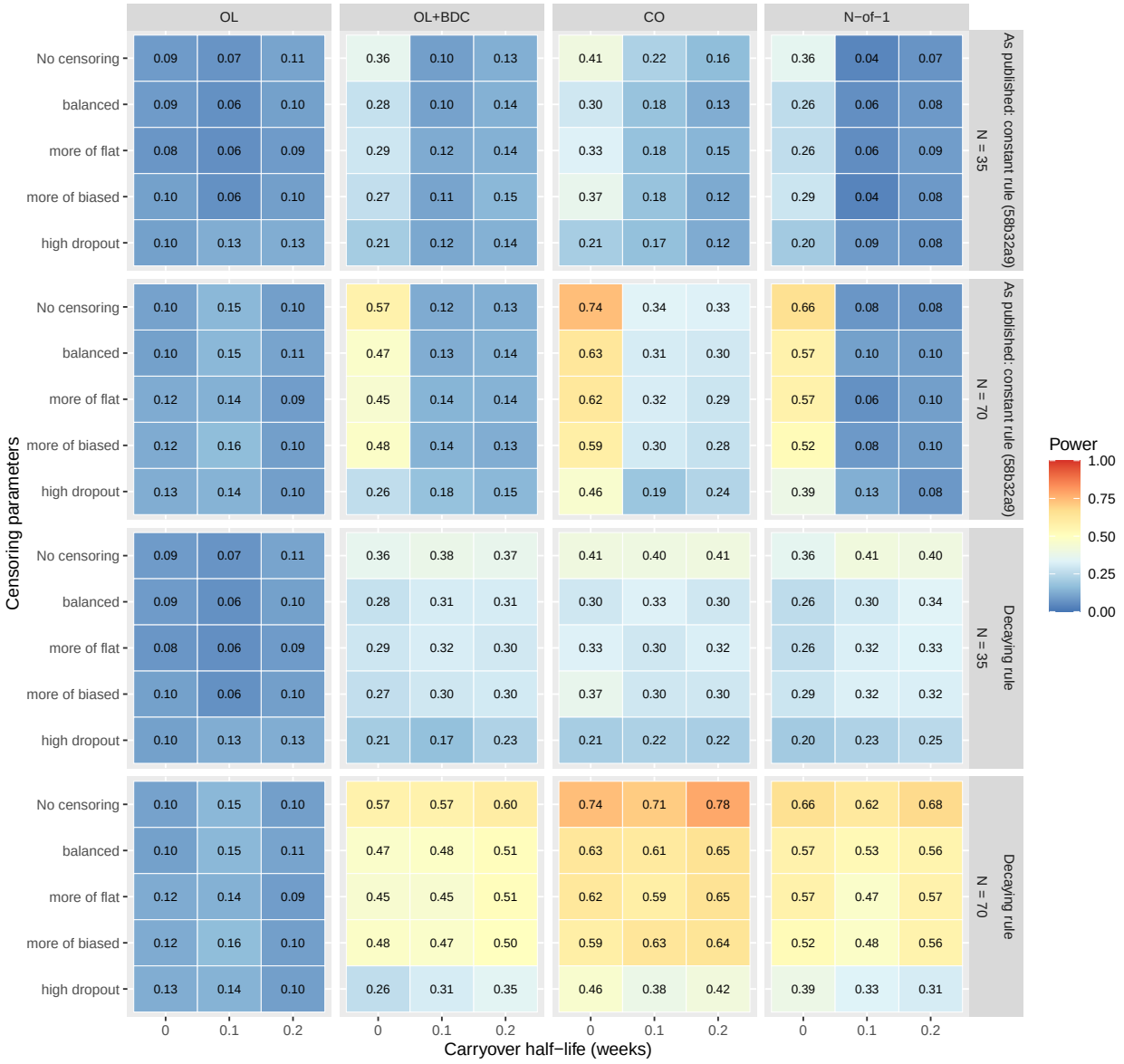


Figure 6: Power to detect the biomarker-by-treatment interaction from the code of Hendrickson et al. at commit 58b32a9, in the layout of their Figure 4B, as published (upper two panel rows) and with the decaying biomarker-response correlation (lower two panel rows). Rows within panels: their censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, binary-indicator analysis, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions. No matrix was repaired in either version.

small demand on the structure. It also carries no interaction (Stage 2). The cells that need no repair under the constant rule are therefore the cells in which the interaction is absent.

Discussion

The correlation encoding of the interaction

Conditional structure. Under joint normality, conditioning on the biomarker shows what the correlation encoding does to the drug response. With b_i the standardized biomarker,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm}\, g_t\, \sigma_{BR}\, b_i, \qquad \mathrm{Cov}(BR_{i\cdot} \mid B_i) = \Sigma_{BR} - c_{bm}^2 \sigma_{BR}^2\, g g^\top.$$

The encoding therefore shifts the conditional drug-response mean in proportion to the correlation profile g_t. It reduces the residual drug-response covariance by a rank-one term. Two consequences follow.

- **The power loss under carryover is set by the profile.** The size of the loss depends on whether the interaction fades with the drug effect, which is the correlation rule.
- **The feasibility bound is the condition** that the reduced residual covariance remains positive definite.

The correlation encoding is therefore a parameterization of the interaction as a correlation. It keeps the marginal variance of the drug response fixed. Beyond its first use (2) we found no use of it in the N-of-1 literature we searched (Background). Its strengths and its limits both follow from it.

The correlation encoding has four strengths.

- **One interpretable parameter.** The interaction strength is a single correlation. Its square, c_{bm}^2, is the share of on-drug drug-response variance the biomarker explains, the systematic part of treatment-effect variation in the sense of Ding et al. (55).
- **A reduced form of responder heterogeneity.** Suppose participants fall into two latent responder classes, and both the biomarker and the drug response differ between classes. Then class membership carries the covariance between them. Matching second moments gives $c_{bm}^2 =$

$f_B f_{BR}$, where f_B and f_{BR} are the fractions of the biomarker's and the drug response's variance that lie between classes (Appendix A.6). The encoding therefore matches the second moments of a responder-class mechanism without requiring the mixture to be identified. Identifying the mixture typically needs larger samples than N-of-1 programs enroll (56–58). The match is limited to second moments. The mixture's conditional mean is nonlinear in the biomarker, and its higher moments differ.

- **The marginal variance of the drug response is fixed.** The biomarker explains part of the existing variation rather than adding to it. The effect size can therefore be stated as a share of response variance without changing the response distribution.
- **It fits a biomarker that marks individual sensitivity to the drug** better than one that sets a fixed dose-response shift.

It also has four structural limits.

- **The joint distribution must exist,** which bounds the effect size. In outcome units the bound is $c^*_{bm}\sigma_{BR}$ per SD of biomarker, 3.8 points under the recommended DGP at the default σ_{BR}. The planning section shows this is below the association first reported for prazosin, which did not replicate in CSP 563. The bound depends on the correlation structure and the design.
- **The correlation must be specified at off-drug visits.** A constant correlation after first exposure removes the interaction under carryover. The specification adopted here is itself asymmetric between titration and washout (Stage 2).
- **The drug-response component keeps its full variance at off-drug visits,** with zero mean. This adds noise to every on-off contrast. It belongs to the component decomposition, not to the correlation encoding.
- **A correlation is less intuitive than a slope for clinical readers.** The slope it implies, $-c_{bm}\sigma_{BR}/\sigma_{bm}$, must be derived before it can be compared with an observed effect. The comparison depends on σ_{BR}, which is not estimated.

A model that generates the drug response from a participant-level sensitivity to exposure would address these limits:

$$BR_{it} = \omega_i f(t) + \varepsilon_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta b_i) + u_i,$$

where $f(t)$ is the exposure-response time course, ζ the biomarker's effect on sensitivity and u_i the sensitivity it does not explain. This is a random-coefficient model with a covariate-by-exposure interaction, the form used to describe individual response in replicated crossover and N-of-1 trials (6,27).

- It encodes the biomarker through individual sensitivity, so it has no feasibility bound.
- Its moderation is proportional to the response at every visit, so it fades during titration as well as after discontinuation.
- Its off-drug variance shrinks with the response.

It is not yet implemented in the software. Its response half-life for prazosin in PTSD is not known. Until then, the correlation encoding's range of use is effect sizes below the feasibility bound, which the planning section translates into outcome units.

We do not claim that the correlation encoding is preferable to the random-coefficient form. Whether the two give the same power, feasibility and design ranking at a matched implied slope has not been tested.

What remains uncalibrated

The recommended DGP meets requirements 1 and 2 by construction and derivation, and requirement 3 through its options. Two of its choices rest on pharmacological judgment: the restart of time on drug at re-exposure and the assumed half-life range. The separable form matches the analysis model's residual structure only within each phase, and the effect of the departure across phases on test calibration has not been measured. The DGP is not calibrated. Every parameter is a default value or a choice made here. The values that matter most are the serial correlation, the drug-response SD and the symptom half-life. The parallel-group trial CSP 563 informs only the first, and only on the total score. Its CAPS correlation decays slowly, and the DGP's total-score correlation decays faster at short lags (planning section). In the reanalysis (Additional file 1), compound symmetry fit it worst among the structures compared. All three values could be estimated from an aggregated N-of-1 trial of prazosin with weekly symptom measurement. Such a trial would also show which correlation form fits weekly series.

Recommendations

For simulation studies of these designs, we recommend the following.

1. **Let the biomarker-response correlation decay after discontinuation,** and stop on matrices that are not positive definite.
2. **State the choices:** carryover decaying from the last on-drug mean, AR(1) in weeks at a reported ρ, and the separable cross-component form. Repeat the analysis at neighboring ρ where the conclusion could depend on it.
3. **Choose c_{bm} below the feasibility bound** for every design in the study. Report it on two scales: as c_{bm}, and as the implied slope in outcome points per SD of biomarker.
4. **Use the sensitivity model, once available, for effects above the feasibility bound.** Until then, state the bound in outcome units as a limitation.

Limitations

- **The closed form covers one statistic.** It is exact for the paired-difference statistic, not for the binary-indicator mixed model. The two agree closely in the Hybrid design. Under compound symmetry, in the crossover design, the closed form understates the binary-indicator mixed model. The sample sizes of the planning table are E9 values. The simulated G10 columns show the direction and size of the difference under the recommended DGP: at the E9 sample size G10 power was 0.844 to 0.936 in the uncapped cells. The G10 sample size for 80 percent power was searched for in the Hybrid design only, by simulation at 6 values of N per cell. It carries Monte Carlo error that Table 9 does not show.
- **The planning results depend on ρ.** The main tables use $\rho = 0.7$. The one available PTSD series suggests a slower decay (Stage 4). At $\rho = 0.9$ the Hybrid design needs about half the sample, the largest feasible effect is lower, and G10 fits fail often in the crossover design (Table 11).
- **The biomarker is assumed not to be prognostic.** Every result assumes a biomarker correlated with the drug response only (Aims). CSP 563 supports the assumption for baseline severity but cannot test it for the placebo response or the natural history. Under a prognostic biomarker some analyses, including G10, lose their nominal size, and the planning sample sizes do not apply (45).

- **At pharmacological half-lives the recommended DGP shows no carryover effect.** The on/off correlation difference is within 0.01 of 1 at 0.2 week. The DGP therefore tests carryover only at half-lives of 0.5 week and above, which we cannot ground pharmacologically.
- **The correlation is asymmetric between titration and washout.** Full moderation during the first weeks on drug is a modeling choice, not a consequence of the pharmacology.
- **Parameter values are not estimated.** The Gompertz parameters, correlations and σ_{BR} are default values or chosen values. A reanalysis of CSP 563 estimates the total-score variance and correlation, which differ from the DGP's, and cannot identify the component split, the expectancy weight or the half-life. The motivating blood-pressure association did not replicate there.
- **The DGP is Gaussian with a single biomarker.**
- **The pharmacological carryover range** for prazosin at weekly visits is inferred from its plasma half-life, not measured as a symptom-response half-life.
- **The stress test is exploratory.** Cells and analyses were added after first results.
- **The literature search is incomplete.** We searched PubMed and Europe PMC, not Web of Science or Scopus, and the searches missed one relevant study that we found by other means (Background). A prior use of the correlation encoding may therefore exist.
- **The correlation encoding is not compared with the alternative.** The random-coefficient form of the Discussion has no feasibility bound and no off-drug drug-response noise. We have not compared the two at a matched implied slope.

Conclusions

We set out to move toward an optimal DGP for simulating predictive-biomarker interactions in aggregated N-of-1 trials. We took its elements one at a time, and three choices matter most. First, the biomarker-response correlation must decay after discontinuation. If it stays constant after first exposure, the interaction is absent among the post-baseline visits under any carryover. Power then falls nearly to the test size. With a decaying biomarker-response correlation, power returns to its no-carryover value at pharmacological half-lives (the on/off correlation difference is within 0.01 of 1 at 0.2 week). It declines smoothly at half-lives of 0.5 week and above. The decline is proved monotone for the path-stratified paired-difference statistic and was checked on a grid for the pooled statistic. Second, the serial correlation structure sets the largest interaction the DGP can hold. A

separable AR(1) structure at a serial correlation of 0.7 allows a biomarker-response correlation of 0.47. That is 3.8 symptom points per SD of biomarker at the default drug-response SD. The value 0.7 is a convention, however, and the one available PTSD series suggests a slower decay. At 0.9 the bound falls to 0.340, and the Hybrid design needs 154 participants at $c_{bm} = 0.25$ instead of 278. Third, a matrix outside the bound should be refused rather than repaired.

The planning table gives the power at matched strength. It does so for the paired-difference statistic in closed form and for the recommended mixed-model analysis G10 by simulation. At the closed-form sample size, G10 power was 0.844 to 0.936 in the uncapped cells. In the Hybrid design the G10 sample size for 80 percent power was 0.71 to 0.83 times the closed-form value. We propose the recommended DGP as a reference for simulation studies of these designs when the biomarker is predictive only. It has options for measurement error, the decay form and dropout; the prognostic case is the subject of a companion paper. Every parameter is labeled as a default value, a choice or uncalibrated. Calibrating the serial correlation, the drug-response SD and the symptom half-life is the next step toward an optimal DGP.

Glossary

Terms are listed in the order in which the paper introduces them. Labels in backticks are the identifiers used in the repository's code, data files and table headings.

- **Recommended DGP** (`refined` in code and data labels): the data-generating process of Table 8: the decaying biomarker-response correlation, the feasibility check, carryover decaying from the last on-drug mean, the separable AR(1) structure and the optional features.
- **Default values** (`published` in code and data labels): the parameter values of Hendrickson et al. (2020), used throughout (Methods).
- **Correlation encoding of the interaction** (`mvn`): the predictive biomarker is a correlation between the biomarker and the drug response that is present on drug and absent off drug.
- **Biomarker-response correlation** (c_{bm}, `c.bm`): the correlation between the baseline biomarker and the drug-response component at an on-drug visit, which sets the strength of the interaction.
- **Correlation profile** (g_t): the fraction of c_{bm} present at visit t, equal to 1 on drug.

- **Constant rule** (`step` in code and data labels): the biomarker-response correlation is c_{bm} at every visit where the drug-response mean is nonzero, and so is constant after first exposure.
- **Decaying rule** (the decaying biomarker-response correlation; `graded` in code and data labels): the correlation is c_{bm} on drug, $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before exposure.
- **On/off correlation difference** ($\mathcal{G}$): the on-drug value of g_t minus the mean of g_t over the post-baseline off-drug visits, $1 - \bar{g}_{\text{off}}$, through which alone the correlation rule affects closed-form power.
- **Feasibility bound** (c^*_{bm}, "Ceiling" in tables): the largest c_{bm} for which the correlation matrix is positive definite under a given correlation structure, design and correlation profile.
- **Carryover half-life** ($t_{1/2}$, `carryover_t1half`): the time after discontinuation for the drug-response mean, and under the decaying rule the biomarker-response correlation, to fall by half.
- **Pharmacological and behavioral half-lives**: half-lives of 0.1 to 0.2 week are the assumed pharmacological range for prazosin, and half-lives of 0.5 week and above are labeled behavioral scenarios.
- **Compounding carryover rule** (`recursive`): the drug-response mean at an off-drug visit is the previous visit's mean decayed by the time since discontinuation, so the decay compounds across visits.
- **Anchored carryover rule** (`anchored`): the drug-response mean at an off-drug visit is the last on-drug mean decayed by the elapsed time since discontinuation.
- **Drug-response component** (BR): the latent response driven by time on drug, with mean on a modified Gompertz curve.
- **Placebo-response component** (PB): the latent response driven by time in the trial and scaled by the expectancy.
- **Natural-history component** (TV): the latent response driven by time in the trial that does not depend on treatment.
- **Baseline severity** (BL): the participant's symptom level before any response, which enters every visit equally.
- **Expectancy** (η_t): the participant's probability of being on drug at visit t given the design, 1 in the open-label phase and 0.5 when blinded.
- **Compound symmetry** (ρ_{cs}): the same correlation between any two visits of a component, whatever their separation.

- **Separable AR(1) structure**: the same first-order autoregressive correlation $\rho^{|w_t - w_s|}$ for every component, with a constant between-component correlation c_1 at every lag, so that the response block is a Kronecker product.
- **Two-constant form**: an AR(1) structure in which the same-visit cross-component correlation c_1 and the other-visit correlation $c_\times \rho^{|w_t - w_s|}$ use different constants.
- **Hybrid design** (`Hybrid`): open-label titration, blinded discontinuation, then a brief crossover, with visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20.
- **OL+BDC** (`OLBDC`): open label followed by blinded discontinuation.
- **CO**: the crossover design, in which participants receive drug and placebo in one of two orders.
- **OL**: the open-label design, in which every visit is on drug.
- **Path**: one sequence of drug and placebo phases within a design, to which participants are allocated in fixed numbers.
- **Paired-difference statistic** (E9, E9b): the per-participant difference between the mean on-drug and mean off-drug scores, regressed on the biomarker, using post-baseline visits only (E9) or counting baseline as one more off-drug visit (E9b).
- **Binary-indicator analysis**: the linear mixed model with a random intercept, a binary drug indicator and baseline included as an off-drug row (Methods). The companion paper calls it E0, because its own analysis G1 also uses a binary indicator.
- **G10** (`E7null_cr2` in data labels): the analysis the companion paper recommends. It is a random-intercept CAR(1) mixed model with the exposure-decayed drug indicator. The half-life is chosen by AIC over 0.25, 0.5, 1 and 2 weeks from maximum-likelihood fits without the interaction. The model is then refitted by REML with the interaction and tested with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom.
- **G9** (`E7cr2` in data labels): as G10, except that the half-life is chosen with the interaction in the model.
- **Working-covariance standard error** (model-based): the standard error computed as if the analysis model's assumed covariance of each participant's outcomes were the true one. Correct only when that covariance is right.
- **Cluster-robust standard error** (CR2, sandwich): the standard error estimated from the residuals with each participant as a cluster, valid when the assumed covariance is wrong; used here in its bias-reduced CR2 form with Satterthwaite degrees of freedom.
- **Capped cell** († in Table 10): a cell whose E9 sample size for 80 percent power exceeds 400,

simulated at $N = 400$.

- **Exposure-decayed drug indicator** (D_{bc}): a drug indicator equal to 1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before exposure.
- **Prognostic biomarker**: a biomarker associated with the outcome through a channel other than the drug response, such as baseline severity, the placebo response or the natural history. This paper assumes the biomarker is not prognostic.
- **Expectancy-term model**: the random-intercept CAR(1) analysis that adds the expectancy and a biomarker-by-expectancy term to the mean model.
- **CAR(1)**: the continuous-time first-order autoregressive residual correlation `corCAR1` of the analysis model.
- **Stress test** (`11-strawman-stress-test.R`): study 2, the simulation that fits fourteen analyses to trials from the recommended DGP, with and without its optional features. Ten analyses are reported. Its cells with a prognostic biomarker are reported in the companion paper.
- **Runs of the stress test** (`summary-full*.csv`, `summary-full*-anchored.csv` and `summary-full*-anchored-fixed-ha0.5.csv`): the first run used the compounding carryover rule (`recursive`) and fixed allocation. The second used carryover decaying from the last on-drug mean (`anchored`) and randomized allocation. The third, reported in Stage 6, used the anchored carryover rule and fixed allocation, the recommended DGP as defined.
- **The three requirements**: (1) a valid joint distribution: the correlation matrix is positive definite, ensured by the separable AR(1) structure and a biomarker-response correlation below the feasibility bound; (2) an interaction that survives carryover: it remains an on-minus-off contrast; (3) flexibility for stress conditions: dropout, measurement error and decay form as options.

Abbreviations

ADEMP: aims, data-generating mechanisms, estimands, methods, performance measures; AIC: Akaike information criterion; AR(1): first-order autoregressive; BDC: blinded discontinuation; BL: baseline severity; BR: drug-response component; CAPS: Clinician-Administered PTSD Scale; CAR(1): continuous-time first-order autoregressive correlation; CI: confidence interval; CO: crossover; CR2: bias-reduced cluster-robust variance estimator; CS: compound symmetry; CSP: Cooperative Studies Program; DGP: data-generating process; E9, E9b: the paired-difference

interaction statistic on post-baseline visits, and with baseline counted as an off-drug visit; MCSE: Monte Carlo standard error; ML: maximum likelihood; OL: open label; OL+BDC: open label followed by blinded discontinuation; PB: placebo-response component; PTSD: post-traumatic stress disorder; REML: restricted maximum likelihood; SD: standard deviation; SE: standard error; TV: natural-history component; VA: Department of Veterans Affairs.

Supplementary material

The additional files are in the `supplement/` directory of this manuscript in the repository. They are project documents, not peer-reviewed publications. Each is cited for the result named here.

- **Additional file 1** (`S1-csp563-reanalysis.pdf`). The reanalysis of CSP 563: the arm-by-blood-pressure interaction, the biomarker's association with baseline severity, and the variance and serial correlation of the total symptom score. Aggregate results only; the participant-level data are not public.
- **Additional file 2** (`S2-stress-test-protocol.pdf`). The working protocol of the stress test (study 2).
- **Additional file 3** (`S3-analysis-selection.pdf`). The post hoc comparison and selection of analyses from the stress test.
- **Additional file 4** (`S4-stress-test-results.pdf`). Stress-test results, including the size-adjusted power comparison of the expectancy-term and CAR(1) models.
- **Additional file 5** (`S5-compendium-mathematics.pdf`). The compendium of mathematical results, including the derivations behind Appendix A.4.
- **Additional file 6** (`S6-separable-covariance.pdf`). The full derivation and numerical check of Appendix A.5.
- **Additional file 7** (`S7-carryover-recursion.pdf`). The carryover recursion in the code of Hendrickson et al. (Appendix B.3).

Declarations

Ethics approval and consent to participate. Not applicable: the study uses simulated data only.

Consent for publication. Not applicable.

Use of CSP 563 results. Additional file 1 reports aggregate results from a reanalysis of VA Cooperative Studies Program trial CSP 563. No participant-level data are shared. [**Placeholder: confirmation that publication of these aggregate results is permitted under the data use agreement is pending.**]

Availability of data and materials. All code and simulated data are in the pmsimstats-ng repository (`https://github.com/rgt47/pmsimstats-ng`). A tagged release with a persistent identifier will be deposited at submission. Until then, cite commit dd6d38686, from which this version was rendered with uncommitted changes; a submitted version must be rendered from a clean commit.

- the closed-form calculations, in `analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-fo` and the serial-correlation stage, planning table and trajectory figure, in `17-rho-stage.R` in the same directory;
- the G10 and G9 simulations at the planned sample sizes, on the Figure 4B grid and for the G10 sample size, in `19-g9-planning-and-fig4b.R` in the same directory (run with `--params published --specs E7null_cr2,E7cr2`, blocks P and F in one job and block N in another), with their outputs under `analysis/data/quick-sim/carryover-closed-form/published-planning/`; the generator check against the closed form in `analysis/scripts/carryover-sensitivity/42-check-para` the earlier runs under `g10-planning/` and `g9-planning/` used other response SDs and are not reported;
- the paired rerun of Appendix B.4, in `analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.` (run with `--params published`), with its output under `analysis/data/carryover-sensitivity/publishe`
- the rerun of the code of Hendrickson et al. (section "Differences from the DGP of Hendrickson et al."), in `analysis/scripts/quick-sim/hendrickson-problems/14-graded-patch-58b32a9.R`, the 4,000-replicate rerun at 0.1 week in `17-published-th01-more-reps.R`, and the repair counts in `11-pd-repair-count.R`;
- the stress test, in `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R`. This is study 2. The reported (third) run is reproduced with `--carryover anchored --allocation fixed --analysis-half-life 0.5`, the second with `--carryover anchored --allocation random --analysis-half-life 0.5`, and the first with `--carryover recursive --allocation fixed --analysis-half-life 0.5`, each followed by the same command with `--extra` and with `--extra2` added. The script's default analysis half-life is

now 0.2 week, so the flag is needed;

- the code of Hendrickson et al., at `github.com/rchendrickson/pmsimstats`, commit 58b32a9.

Competing interests. The authors declare no competing interests.

References

1. Sargent DJ, Conley BA, Allegra C, Collette L. Clinical trial designs for predictive marker validation in cancer treatment trials. Journal of Clinical Oncology. 2005;23(9):2020–7. doi:10.1200/JCO.2005.01.112

2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. Frontiers in Digital Health. 2020;2:13. doi:10.3389/fdgth.2020.00013

3. Nikles J, Mitchell GK, Schluter P, Good P, Hardy J, Rowett D, et al. Aggregating single patient (n-of-1) trials in populations where recruitment and retention was difficult: The case of palliative care. Journal of Clinical Epidemiology. 2011;64(5):471–80. doi:10.1016/j.jclinepi.2010.05.009

4. Vohra S, Shamseer L, Sampson M, Bukutu C, Schmid CH, Tate R, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. Journal of Clinical Epidemiology. 2016;76:9–17. doi:10.1016/j.jclinepi.2015.05.004

5. Punja S, Xu D, Schmid CH, Hartling L, Urichuk L, Nikles CJ, et al. N-of-1 trials can be aggregated to generate group mean treatment effects: A systematic review and meta-analysis. Journal of Clinical Epidemiology. 2016;76:65–75. doi:10.1016/j.jclinepi.2016.03.026

6. Senn S. Mastering variation: Variance components and personalised medicine. Statistics in Medicine. 2016;35(7):966–77. doi:10.1002/sim.6739

7. Schork NJ. Personalized medicine: Time for one-person trials. Nature. 2015;520(7549):609–11. doi:10.1038/520609a

8. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. American Journal of Psychiatry. 2003;160(2):371–3. doi:10.1176/appi.ajp.160.2.371

9. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. American Journal of Psychiatry. 2013;170(9):1003–10. doi:10.1176/appi.ajp.2013.12081133

10. Raskind MA, Peskind ER, Chow B, et al. Trial of prazosin for post-traumatic stress disorder in military veterans. New England Journal of Medicine. 2018;378(6):507–17. doi:10.1056/NEJMoa1507598

11. Raskind MA, Millard SP, Petrie EC, Peterson K, Williams T, Hoff DJ, et al. Higher pretreatment blood pressure is associated with greater posttraumatic stress disorder symptom reduction in soldiers treated with prazosin. Biological Psychiatry. 2016;80(10):736–42. doi:10.1016/j.biopsych.2016.03.2108

12. Duan N, Norman D, Schmid C, Sim I, Kravitz RL. Personalized data science and personalized (N-of-1) trials: Promising paradigms for individualized health care. Harvard Data Science Review. 2022;(Special Issue 3). doi:10.1162/99608f92.8439a336

13. Schmid CH, Yang J. Bayesian models for N-of-1 trials. Harvard Data Science Review. 2022;(Special Issue 3). doi:10.1162/99608f92.3f1772ce

14. Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. Healthcare. 2019;7(3):84. doi:10.3390/healthcare7030084

15. Senn S. Cross-over trials in clinical research. 2nd ed. Chichester, UK: John Wiley & Sons; 2002.

16. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. Statistics in Medicine. 2019;38(11):2074–102. doi:10.1002/sim.8086

17. Percha B, Baskerville EB, Johnson M, Dudley JT, Zimmerman N. Designing robust N-of-1 studies for precision medicine: Simulation study and design recommendations. Journal of Medical Internet Research. 2019;21(4):e12641. doi:10.2196/12641

18. Chen X, Chen P. A comparison of four methods for the analysis of N-of-1 trials. PLoS ONE. 2014;9(2):e87752. doi:10.1371/journal.pone.0087752

19. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. Healthcare. 2019;7(4):137. doi:10.3390/healthcare7040137

20. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. BMC Medical Research Methodology. 2023;23(1):191. doi:10.1186/s12874-023-02012-5

21. Tueller S, Ramirez D, Cance JD, Ye A, Wheeler AC, Fan Z, et al. Power analysis for idiographic (within-subject) clinical trials: Implications for treatments of rare conditions and precision medicine. Behavior Research Methods. 2023;55(8):4175–99. doi:10.3758/s13428-022-02012-1

22. Grenet G, Blanc C, Bardel C, Gueyffier F, Roy P. Comparison of crossover and parallel-group designs for the identification of a binary predictive biomarker of the treatment effect. Basic and Clinical Pharmacology and Toxicology. 2020;126(1):59–64. doi:10.1111/bcpt.13293

1359 23. Van den Noortgate W, Onghena P. Hierarchical linear models for the quantitative integration of effect sizes in single-case research. Behavior Research Methods, Instruments, and Computers. 2003;35(1):1–10. doi:10.3758/BF03195492

1360 24. Moeyaert M, Ferron JM, Beretvas SN, Van den Noortgate W. From a single-level analysis to a multilevel analysis of single-case experimental designs. Journal of School Psychology. 2014;52(2):191–211. doi:10.1016/j.jsp.2013.11.003

1361 25. Moeyaert M, Yang P, Xu X. The power to explain variability in intervention effectiveness in single-case research using hierarchical linear modeling. Perspectives on Behavior Science. 2022;45(1):13–35. doi:10.1007/s40614-021-00304-z

1362 26. Senn S. Individual response to treatment: Is it a valid assumption? BMJ. 2004;329(7472):966–8.

1363 27. Araujo A, Julious S, Senn S. Understanding variation in sets of N-of-1 trials. PLoS ONE. 2016;11(12):e0167167. doi:10.1371/journal.pone.0167167

1364 28. Senn S, Rolfe K, Julious SA. Investigating variability in patient response to treatment: A case study from a replicate cross-over study. Statistical Methods in Medical Research. 2011;20(6):657–66. doi:10.1177/0962280210379174

1365 29. Senn S. Sample size considerations for n-of-1 trials. Statistical Methods in Medical Research. 2019;28(2):372–83. doi:10.1177/0962280217726801

1366 30. Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. Journal of Clinical Epidemiology. 1997;50(4):401–10. doi:10.1016/S0895-4356(96)00429-5

1367 31. Kraemer HC, Wilson GT, Fairburn CG, Agras WS. Mediators and moderators of treatment effects in randomized clinical trials. Archives of General Psychiatry. 2002;59(10):877–83.

1368 32. Ghosh S, Henderson SG. Behavior of the NORTA method for correlated random vector generation as the dimension increases. ACM Transactions on Modeling and Computer Simulation. 2003;13(3):276–94.

1369 33. Joe H. Generating random correlation matrices based on partial correlations. Journal of Multivariate Analysis. 2006;97(10):2177–89. doi:10.1016/j.jmva.2005.05.010

1370 34. Jones B, Kenward MG. Design and analysis of cross-over trials. 3rd ed. Boca Raton, FL: CRC Press; 2014.

1371 35. Senn SJ. Is the "simple carry-over" model useful? Statistics in Medicine. 1992;11(6):715–26. doi:10.1002/sim.4780110603

1372 36. Senn S, Lambrou D. Robust and realistic approaches to carry-over. Statistics in Medicine. 1998;17(24):2849–64.

1373 37. Holford N. Clinical pharmacology = disease progression + drug action. British Journal of Clinical Pharmacology. 2015;79(1):18–27. doi:10.1111/bcp.12170

1374 38. Galecki AT. General class of covariance structures for two or more repeated factors in longitudinal data analysis. Communications in Statistics - Theory and Methods. 1994;23(11):3105–19. doi:10.1080/03610929408831436

1375 39. Naik DN, Rao SS. Analysis of multivariate repeated measures data with a Kronecker product structured covariance matrix. Journal of Applied Statistics. 2001;28(1):91–105. doi:10.1080/02664760120011626

1376 40. Li F, Chen X, Tian Z, Esserman D, Heagerty PJ, Wang R. Designing three-level cluster randomized trials to assess treatment effect heterogeneity. Biostatistics. 2023;24(4):833–49. doi:10.1093/biostatistics/kxac026

1377 41. Hobbs DC, Twomey TM, Palmer RF. Pharmacokinetics of prazosin in man. Journal of Clinical Pharmacology. 1978;18(8-9):402–6. doi:10.1002/j.1552-4604.1978.tb02456.x

1378 42. Holford NHG, Sheiner LB. Understanding the dose-effect relationship: Clinical application of pharmacokinetic-pharmacodynamic models. Clinical Pharmacokinetics. 1981;6(6):429–53. doi:10.2165/00003088-198106060-00002

1379 43. Haller B, Ulm K. A simulation study on estimating biomarker-treatment interaction effects in randomized trials with prognostic variables. Trials. 2018;19(1):128.

1380 44. Kent DM, Paulus JK, Klaveren D van, et al. The predictive approaches to treatment effect heterogeneity (PATH) statement. Annals of Internal Medicine. 2020;172(1):35–45. doi:10.7326/M18-3667

1381 45. pmsimstats team. When the biomarker is also prognostic: Bias in the biomarker-treatment interaction test of aggregated N-of-1 trials and two remedies. 2026.

1382 46. Kuznetsova A, Brockhoff PB, Christensen RHB. lmerTest package: Tests in linear mixed effects models. Journal of Statistical Software. 2017;82(13):1–26. doi:10.18637/jss.v082.i13

1383 47. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. Survey Methodology. 2002;28(2):169–81.

1384 48. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. Journal of Business and Economic Statistics. 2018;36(4):672–83. doi:10.1080/07350015.2016.1247004

1385 49. Siepe BS, Bartoš F, Morris TP, Boulesteix AL, Heck DW, Pawel S. Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. Psychological Methods. 2024. doi:10.1037/met0000695

1386 50. pmsimstats team. Carryover representation and inference for biomarker-treatment interaction tests with a predictive biomarker in aggregated N-of-1 trials under covariance moderation: A staged simulation study and a recommended analysis for the Hybrid design. 2026.

1387 51. Pawel S, Kook L, Reeve K. Pitfalls and potentials in simulation studies: Questionable research practices in comparative simulation studies allow for spurious claims of superiority of any method. Biometrical Journal. 2024;66(1):2200091. doi:10.1002/bimj.202200091

1388 52. Higham NJ. Computing a nearest symmetric positive semidefinite matrix. Linear Algebra and its Applications. 1988;103:103–18. doi:10.1016/0024-3795(88)90223-6

1389 53. Schork NJ. Accommodating Serial Correlation and Sequential Design Elements in Personalized Studies and Aggregated Personalized Studies. Harvard Data Science Review. 2022;(Special Issue 3). doi:10.1162/99608f92.f1eef6f4

1390 54. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. Biometrics. 1997;53(3):983–97. doi:10.2307/2533558

1391 55. Ding P, Feller A, Miratrix L. Decomposing treatment effect variation. Journal of the American Statistical Association. 2019;114(525):304–17. doi:10.1080/01621459.2017.1407322

1392 56. Bauer DJ, Curran PJ. Distributional assumptions of growth mixture models: Implications for overextraction of latent trajectory classes. Psychological Methods. 2003;8(3):338–63. doi:10.1037/1082-989X.8.3.338

1393 57. Lubke GH, Muthén B. Investigating population heterogeneity with factor mixture models. Psychological Methods. 2005;10(1):21–39. doi:10.1037/1082-989X.10.1.21

1394 58. Nylund KL, Asparouhov T, Muthén BO. Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. Structural Equation Modeling. 2007;14(4):535–69. doi:10.1080/10705510701575396

1395 59. Higham NJ, Strabić N, Šego V. Restoring definiteness via shrinking, with an application to correlation matrices with a fixed block. SIAM Review. 2016;58(2):245–63. doi:10.1137/140996112

Appendix A. Closed-form derivations

A.1 Moments of the paired-difference statistic

Let $a_t = 1/n_{\text{on}}$ at on-drug visits and $-1/n_{\text{off}}$ at off-drug visits, so that $\Delta_i = \sum_t a_t Y_{it}$. Because the weights sum to zero, the baseline level cancels and $\Delta = -\sum_t a_t S_t$, with $S_t = TV_t + PB_t + BR_t$. In general $\text{Var}(\Delta) = a^\top \Sigma_S a$, where Σ_S is the covariance of the summed response, whatever the structure.

With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_\eta = \sum a_t \eta_t$, $m_{\eta\eta} = \sum a_t^2 \eta_t^2$ and $m_{2\eta} = \sum a_t^2 \eta_t$, with η_t the design expectancy, the variance under compound symmetry with common correlation ρ_{cs} is

$$\begin{aligned}\text{Var}(\Delta) = {} & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho_{cs} m_1^2 + (1-\rho_{cs}) m_2] + \sigma_{PB}^2 [\rho_{cs} m_\eta^2 + (1-\rho_{cs}) m_{\eta\eta}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_\times m_1^2 + (c_1 - c_\times) m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_\times m_1 m_\eta + (c_1 - c_\times) m_{2\eta}],\end{aligned}$$

where c_1 and $c_\times$ are the same-visit and other-visit cross-component correlations. Under the separable form of Stage 5, $\text{Cov}(S_t, S_s) = (s_t^\top K s_s) A_{ts}$ (A.5), and $\text{Var}(\Delta) = \sum_{t,s} a_t a_s (s_t^\top K s_s) A_{ts}$, with s_t the vector of component standard deviations at visit t.

- **Proposition 1.** No term depends on $t_{1/2}$ under either structure, so $\text{Var}(\Delta)$ is free of the half-life. The argument applies to any correlation matrix in which the half-life does not enter the response block. It therefore holds for unrepaired matrices only. The repair changes the response block, and its changes depend on the half-life through the correlation profile. For E9 the weights sum to zero, so $m_1 = 0$. For E9b the post-baseline weights sum to $1/(n_{\text{off}} + 1)$, and the baseline row, with $S_0 = 0$, carries the remaining weight.
- **Proposition 2.** Only BR_t is correlated with B, with covariance $c_{bm} g_t \sigma_{bm} \sigma_{BR}$, so $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm}) \sum_t a_t g_t$. With $g_t = 1$ on drug, this gives the expressions in the Methods.
- **Residual variance.** Under the correlation encoding, $\sigma_\Delta^2 = \text{Var}(\Delta) - \beta_{bm:D}^2 \sigma_{bm}^2$.

A.2 Power

Within a path of n participants, the least-squares slope is exactly normal given the biomarker values. Its t-statistic is exactly noncentral t with $n-2$ degrees of freedom and noncentrality $\beta_{bm:D}\sqrt{\text{SS}_B}/\sigma_\Delta$,

where $\mathrm{SS}_B \sim \sigma^2_{bm}\chi^2_{n-1}$.

At $\mathrm{SS}_B = \sigma^2_{bm}(n-1)$ the noncentrality is $\Lambda = \sqrt{n-1}\, r_{\Delta B}/\sqrt{1-r^2_{\Delta B}}$, where $r_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\mathrm{SD}(\Delta)$ is proportional to the on/off correlation difference. For small $r_{\Delta B}$ the noncentrality is approximately $\sqrt{n-1}\, r_{\Delta B}$. The sample size for a fixed noncentrality therefore scales as $1/r^2_{\Delta B}$ and hence approximately as $1/\mathcal{G}^2$. The exact scaling uses $r/\sqrt{1-r^2}$.

Pooling paths with a common intercept adds the between-path variation in mean contrast to the residual variance. In the Hybrid design it is under 0.7% of the total. Power is the noncentral-t tail probability averaged over the χ^2 law of SS_B. For the pooled statistic this is an approximation. It treats the between-path differences in mean and slope as extra residual variance and ignores the differences in residual variance between paths. The per-path result is exact.

A.3 Monotonicity under the decaying rule

$$\frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}}\, t^2_{1/2}} \sum_{t\,\text{off}} t_{sd,t}\, 2^{-t_{sd,t}/t_{1/2}} > 0$$

on any path with at least one off-drug visit after an exposure.

- **Proposition 3.** On such a path, by Proposition 2 the slope's magnitude is strictly decreasing in $t_{1/2}$. By A.1, σ^2_Δ is strictly increasing, since $\mathrm{Var}(\Delta)$ is fixed. So the noncentrality and the power of the path-stratified statistic are strictly decreasing. On a path whose off-drug visits all precede first exposure (the placebo-first crossover path), $\bar{g}_{\text{off}} = 0$ at every half-life and the power is constant. The pooled statistic's residual variance also contains the between-path variation of mean contrasts, which depends on the half-life through the carryover means. Its monotonicity was checked on a grid (Stage 2), not proved.
- **Small half-lives.** $\bar{g}_{\text{off}} \le 2^{-1/t_{1/2}}$, which gives the behavior near zero stated in the Results.
- **Large half-lives.** $\bar{g}_{\text{off}} \to 1$ as $t_{1/2} \to \infty$, which recovers the constant rule.

A.4 The feasibility bound

The feasibility bound. Order the variables as biomarker, then responses, and partition the correlation matrix as $\begin{pmatrix} 1 & c_{bm}\tilde{g}^\top \\ c_{bm}\tilde{g} & M \end{pmatrix}$. By the Schur complement, the matrix is positive definite if and only if M is positive definite and $1 - c^2_{bm}\tilde{g}^\top M^{-1}\tilde{g} > 0$. Hence the bound is $c^*_{bm} = (\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$.

A.5 The separable structure

Let $A_{ts} = \rho^{|w_t - w_s|}$ and $K = (1 - c_1)I_3 + c_1 J_3$, and set $M = K \otimes A$, so that the correlation between component c at visit t and component c' at visit s is $K_{cc'}A_{ts}$.

Validity. The eigenvalues of M are the products of those of K and A.

- K has eigenvalues $1 + 2c_1$ and $1 - c_1$ (twice), so it is positive definite for $-\frac{1}{2} < c_1 < 1$.
- A is the correlation matrix of a stationary Ornstein-Uhlenbeck process at the visit times, and is positive definite for any distinct times.

Hence M is positive definite for every schedule. Generatively, $\mathbf{X} = L_A \mathbf{Z} L_K^\top$ with $L_A L_A^\top = A$ and $L_K L_K^\top = K$: each column of $L_A \mathbf{Z}$ is an independent AR(1) path, and the components are fixed mixtures of them.

The feasibility bound. With the biomarker correlated with the drug-response component only, its correlation vector is $c_{bm}\tilde{g}$, with $\tilde{g} = e_3 \otimes g$ and g the correlation profile. Since $M^{-1} = K^{-1} \otimes A^{-1}$,

$$\tilde{g}^\top M^{-1} \tilde{g} = [K^{-1}]_{33}\, g^\top A^{-1} g, \qquad [K^{-1}]_{33} = \frac{1 + c_1}{(1 - c_1)(1 + 2c_1)} = \frac{1}{1 - R^2}.$$

The schedule term. The sampled process is Markov, with step correlation $\rho_t = \rho^{w_t - w_{t-1}}$, so

$$g^\top A^{-1} g = g_1^2 + \sum_{t=2}^{n} \frac{(g_t - \rho_t g_{t-1})^2}{1 - \rho_t^2}.$$

For a binary correlation profile, each step from visit $t-1$ to visit t contributes the following term:

Step	Term
on to on	$(1 - \rho_t)/(1 + \rho_t)$
on to off	$\rho_t^2/(1 - \rho_t^2)$
off to on	$1/(1 - \rho_t^2)$
off to off	0

Under the decaying rule, the off-drug correlation decays instead of switching, which lowers the on-to-off term. This is why the feasibility bound under the decaying rule rises with the half-life. Under

compound symmetry it rises from 0.256 without carryover to 0.307 at 1 week and 0.369 at 2 weeks, which makes $c_{bm} = 0.3$ feasible from 1 week (Figure 4).

The two-constant form. With a common ρ, setting the same-visit cross-component correlation to c_1 and the different-visit correlation to $c_\times A_{ts}$ gives

$$M = K_a \otimes A + K_b \otimes I_n, \qquad K_a = (1 - c_\times) I_3 + c_\times J_3, \qquad K_b = (c_1 - c_\times)(J_3 - I_3).$$

K_b has zero diagonal and so is indefinite. The eigenvalues of M are $\omega_j(A)(1 + 2c_\times) + 2(c_1 - c_\times)$ and $\omega_j(A)(1 - c_\times) - (c_1 - c_\times)$, where $\omega_j(A)$ are the eigenvalues of A. Hence M is positive definite if and only if $\omega_{\min}(A) > (c_1 - c_\times)/(1 - c_\times)$. For visits at a common spacing of s weeks, $\omega_{\min}(A)$ decreases toward $(1 - \rho^s)/(1 + \rho^s)$ as visits accumulate, which is 0.111 at weekly spacing and $\rho = 0.8$ (the threshold itself) and 0.016 at daily spacing. The values for the paper's schedules are in Stage 5.

The compound-symmetry structure. The compound-symmetry form is $M = K_o \otimes I_n + K_p \otimes J_n$, with a person-level factor $K_p = \rho_{cs} I_3 + c_\times (J_3 - I_3)$ and an occasion-level factor $K_o = (1 - \rho_{cs}) I_3 + (c_1 - c_\times)(J_3 - I_3)$. It is positive definite if and only if $(1 - \rho_{cs}) - (c_1 - c_\times) > 0$, whatever the schedule (0.10 at $\rho_{cs} = 0.8$, $c_1 = 0.2$ and $c_\times = 0.1$, which is also the smallest eigenvalue of the eight-visit response block under those values).

The summed response. With component standard deviations s_t and $S_t = \sum_c s_{c,t} X_{c,t}$, separability gives

$$\mathrm{Cov}(S_t, S_s) = (s_t^\top K s_s)\, A_{ts},$$

proportional to the AR(1) kernel when the standard deviations are constant. The correlation of the summed response is

$$\mathrm{Corr}(S_t, S_s) = q_{ts}\, A_{ts}, \qquad q_{ts} = \frac{s_t^\top K s_s}{\sqrt{(s_t^\top K s_t)(s_s^\top K s_s)}} \le 1,$$

by the Cauchy-Schwarz inequality in the inner product defined by the positive definite K. Equality holds if and only if s_s is a multiple of s_t. In the Hybrid design $s_t = (\sigma_{TV}, \sigma_{PB}\eta_t, \sigma_{BR}) = (10, 10\eta_t, 8)$. Within a phase η_t is constant, so $q_{ts} = 1$ and the summed response is exactly AR(1) there. Between an open-label visit ($\eta_t = 1$) and a blinded visit ($\eta_s = 0.5$), with $c_1 = 0.2$, $s_t^\top K s_t = 368$, $s_s^\top K s_s = 257$, $s_t^\top K s_s = 300$ and $q_{ts} = 0.976$. The summed response therefore has a variance that changes at

blinding and a cross-phase correlation 2.4% below the AR(1) value. A constant-variance `corCAR1` model omits both. The analyses with phase-specific residual variances in the stress test represent the first and not the second. The two-constant form adds $(s_t^\top K_b s_t)\,\mathbf{1}[t = s] > 0$, a white-noise term.

These results are derived in full, and checked numerically, in the pmsimstats-ng documentation (Additional file 6).

A.6 The responder-class reading of c_{bm}

Let $Z_i \in \{0, 1\}$ be a latent responder class with $\Pr(Z_i = 1) = p$, and let $B_i = \mu_B(Z_i) + e_{B,i}$ and $BR_i = \mu_{BR}(Z_i) + e_{BR,i}$ for the on-drug drug response, with e_B and e_{BR} independent of each other and of Z. Then $\mathrm{Cov}(B, BR) = \mathrm{Cov}(\mu_B(Z), \mu_{BR}(Z))$. Both $\mu_B(Z)$ and $\mu_{BR}(Z)$ are affine in the binary Z, so their correlation is ± 1 and $|\mathrm{Cov}(B, BR)| = \mathrm{SD}(\mu_B(Z))\,\mathrm{SD}(\mu_{BR}(Z))$. Dividing by $\sigma_B \sigma_{BR}$,

$$c_{bm}^2 = \frac{\mathrm{Var}(\mu_B(Z))}{\sigma_B^2} \cdot \frac{\mathrm{Var}(\mu_{BR}(Z))}{\sigma_{BR}^2} = f_B\, f_{BR},$$

the product of the between-class variance fractions. With more than two classes the correlation of $\mu_B(Z)$ and $\mu_{BR}(Z)$ is at most 1 in magnitude, so $c_{bm}^2 \le f_B f_{BR}$.

Appendix B. Implementation

B.1 to B.3 give the code of three of the changes listed in "Differences from the DGP of Hendrickson et al.", as patches to their code at commit 58b32a9. B.4 and B.5 describe the implementation in the pmsimstats-ng package.

B.1 The decaying biomarker-response correlation

The code of Hendrickson et al. (`R/generateData.R`, lines 131 to 138) set the biomarker-response correlation to c_{bm} wherever the drug-response mean is nonzero:

```
  # correlation with biomarker
  for(p in 1:nP){
```

```
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1,'bm']<-modelparam$c.bm
    correlations['bm',n1]<-modelparam$c.bm
  }
}
```

The replacement sets the correlation in proportion to the remaining drug effect, with zero correlation before first exposure:

```
# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (decaying rule), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(d[p]$tod>0){
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
  }else{
    g<-0
  }
  correlations[n1,'bm']<-modelparam$c.bm*g
  correlations['bm',n1]<-modelparam$c.bm*g
}
```

The constant rule tests whether the off-drug drug-response mean is nonzero. In double precision that mean underflows to zero at late off-drug visits when the half-life is below about 0.007 week, or below about 0.012 week under the compounding carryover rule. The constant rule then sets $g_t = 0$ at those visits. This is a property of the floating-point arithmetic, not of the DGP. It is why Stage 2 describes power under the constant rule as constant at every positive half-life of practical interest.

B.2 Stop instead of repairing

The code of Hendrickson et al. replaces a non-positive-definite covariance matrix with a nearby valid one when its `makePositiveDefinite` option is on, as it is in their driver (lines 145 to 150). It calls `corpcor::make.positive.definite` with tolerance 10^{-3}. That function follows Higham's nearest symmetric positive semidefinite matrix (52). It raises every eigenvalue below 2×10^{-3} to that value, so it changes the variances as well as the correlations. No warning is issued. The replacement below stops instead, so that an infeasible cell is reported rather than changed:

```
# published:
# if(makePositiveDefinite){
#   if(!is.positive.definite(sigma)){
#     sigma<-make.positive.definite(sigma, tol=1e-3)
#   }
# }
if(!corpcor::is.positive.definite(sigma)){
  stop(sprintf('c.bm = %.3f exceeds the feasible range for this design',
               modelparam$c.bm))
}
```

One case is exempt from the stop. The placebo-response component's standard deviation is proportional to the expectancy. A visit with expectancy zero (a run-in or a baseline visit coded with no expectancy) has exactly zero placebo-response variance. It makes Σ singular by construction rather than by an infeasible correlation. The package's stop path checks and factors only the sub-block of positive-variance components and draws the zero-variance components at their means. No correlation is modified. The repair set those variances to 10^{-3} instead.

A repair need not be the unstructured nearest matrix. A shrinking repair can keep the response block fixed and change only the biomarker row (59). Such a repair simulates a smaller c_{bm} than the one requested. The same feasibility question arises for target correlations in NORTA generation (32). Correlation matrices built from partial correlations are positive definite by construction (33), but they do not fix the target correlations a design requires. The stop is the simpler choice. It reports the infeasibility to the user.

The case that fixed the default came from the companion carryover study (50). Two runs combined

Table 14: Correlation matrices that are not positive definite under the recommended DGP (decaying biomarker-response correlation, separable AR(1)). One matrix per design path and half-life: open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The counts are the same at $\rho = 0.7$ (the recommended value) and $\rho = 0.8$ because both feasibility bounds lie between 0.3 and 0.6.

c_{bm} (number of matrices)	Recommended DGP, $\rho = 0.8$	Recommended DGP, $\rho = 0.7$
$c_{bm} = 0.25$ (40)	0 (0%)	0 (0%)
$c_{bm} = 0.3$ (40)	0 (0%)	0 (0%)
$c_{bm} = 0.6$ (40)	30 (75%)	30 (75%)

$c_{bm} = 0.45$ with a second correlation in the biomarker row that made the matrix infeasible at any c_{bm}. The repair lowered the on-drug correlation, and the runs reported the attenuated interaction as a loss of power. A repair can therefore turn an infeasible input into a plausible but wrong finding. The case is documented in the companion paper on prognostic biomarkers (45).

B.3 Carryover decaying from the last on-drug mean

The compounding carryover rule (lines 83 to 92) decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. The replacement decays from the last on-drug value. Study 1 uses it at every positive half-life. Replacing it with the compounding rule changes no slope, variance or feasibility bound, only the path means. It therefore changes the study 1 closed-form power by at most 0.002 in the Hybrid design. With the binary-indicator analysis the change alone moves power by at most 0.006 (Additional file 7, Section 8.3). Combined with the restart of the Gompertz clock at re-exposure, it can produce a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve's value. On the Hybrid schedule it does not (Stage 3).

```
brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
if(nP>1){
  last_on<-0
  for(p in 1:nP){
    if(d[p]$onDrug){
      last_on<-brmeans[p]
    }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
```

```
        brmeans[p]<-last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
      }
    }
  }
```

B.4 The exposure-decayed drug indicator in the analysis

Analyses that use an exposure-decayed drug indicator must code off-drug visits before first exposure as unexposed. The time since discontinuation is zero there, and the decay function returns 1 at zero, so without a guard a placebo-first crossover participant would be coded fully exposed at every visit. In the pmsimstats-ng carryover manuscript this guard was missing. We reran that manuscript's crossover cells in pairs (`analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.R`). The rerun used the default values, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week and 500 replicates, tested with the working-covariance standard error and the analysis half-life equal to the true one. Restoring the guard raised the power of the exposure-weighted analysis from 0.302 to 0.534, within Monte Carlo error of the binary indicator (0.536):

```
exposure_weight <- function(Db, tsd, halflife) {
  w <- 0.5^(tsd / halflife)
  ifelse(Db == 1, 1, ifelse(tsd > 0, w, 0))
}
```

B.5 The recommended DGP in pmsimstats-ng

The package function `buildSigma()` (`R/generateData.R`) implements the decaying rule, with `c.bm` on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure and zero before exposure. It implements AR(1) within each component in calendar weeks. It decays carryover on the drug-response mean from the last on-drug value (not the constant rule of B.3). It also implements prognostic pathways (`c.bm.pb`, `c.bm.bl`, `c.bm.tv`, `b.bm.tv`). They are zero by default, which is the assumption of this paper, and are described in the companion paper (45).

The cross-component block. `buildSigma()` builds the cross-component correlation from two constants: `c.cf1t` at the same visit and `c.cfct * rho^tg` at a lag of `tg` weeks. With `c.cf1t =`

`c.cfct = c_1` this is exactly the separable form of Table 8, $c_1\rho^{|w_t - w_s|}$ at every lag, since $\rho^0 = 1$. With the two constants unequal it is the two-constant form of Stage 5. The recommended DGP is therefore the package with both constants set to 0.2, which is how the project's carryover driver calls it (`analysis/scripts/carryover-sensitivity/simulation-core.R`). Study 2's driver builds the separable block directly (`C1 * A` in `11-strawman-stress-test.R`).

Allocation and carryover in study 2. The driver takes `--allocation fixed|random` and `--carryover anchored|recursive`. The first run used fixed allocation (18, 18, 17, 17) and the compounding carryover rule (`recursive`). The second run used carryover decaying from the last on-drug mean (`anchored`) and randomized allocation. The third run used the decaying rule and fixed allocation, the recommended DGP as defined. All three used the same seed rule and an analysis half-life of 0.5 week. The third run changes one setting relative to each earlier run, so the effects of the carryover rule and of the allocation can be read separately. The tables in Stage 6 are from the third run. Randomized allocation is a driver option that is not part of the definition.

Stop instead of repair. `buildSigma()`, `generateData()` and `generateSimulatedResults()` accept a `stopOnNonPD` argument, `TRUE` by default. A non-positive-definite covariance matrix then raises an error, exactly as in B.2, both at the explicit positive-definiteness check and in the Cholesky-factorization fallback that previously repaired regardless of the `makePositiveDefinite` setting. A repair is reproduced only by passing `stopOnNonPD = FALSE` and setting `options(pmsimstats.allow_repair = TRUE)` explicitly, as the scripts that study the repair itself do. Without the option, the repair is locked off: a draw from a matrix that is not positive definite stops with an error whatever the other arguments are. The argument was first added with a `FALSE` default for backward compatibility. The default was changed after two runs of the companion carryover study repaired infeasible cells and reported the attenuated interaction as a power loss.

Not in the package. The measurement-error, Weibull-decay and dropout options of Table 8 are implemented in the stress-test driver (`11-strawman-stress-test.R`) and in the companion study's driver core, and not yet in the package. The package's dropout mechanisms (`censordata.R`) are censoring patterns in which the biased term weights each participant by 100 minus the improvement from baseline, floored at zero and raised to the power `eb1`, so that dropout rises with worsening. Figure 3 uses them.